Module 3

CHY1009 – Chemistry and Environmental Studies

Electrochemistry
and
Chemical Sensors

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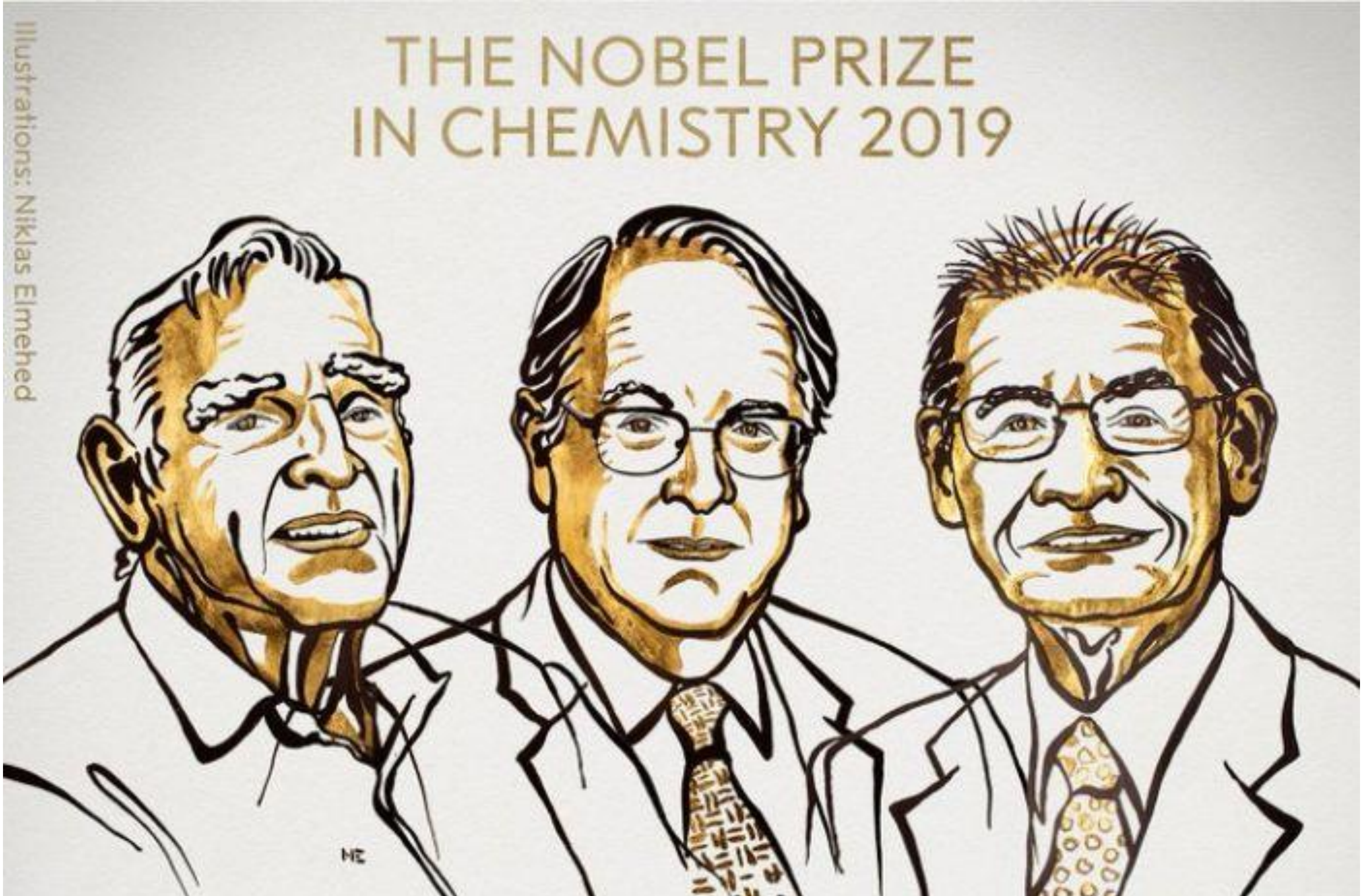
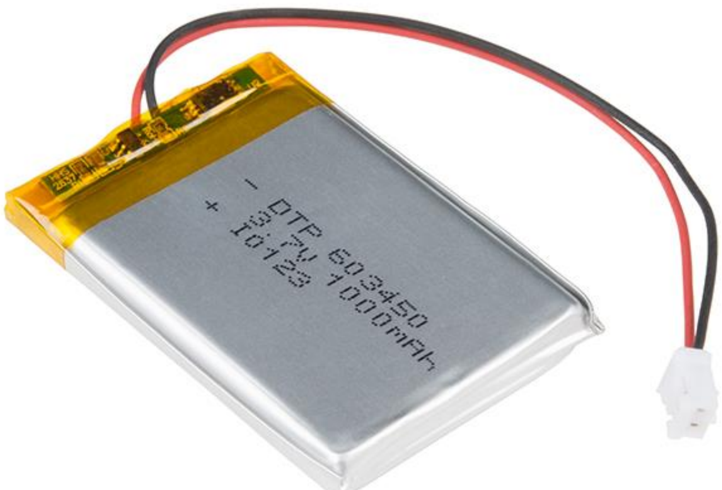
Module 3

Electrochemistry and Chemical Sensors

- Introduction to the electrochemistry
- Cell potential
- Batteries: Primary and secondary batteries
- Corrosion
- Chemical Sensors

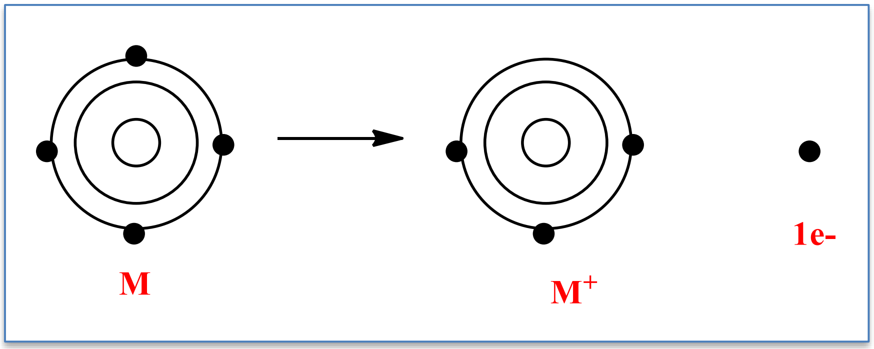

THE NOBEL PRIZE
IN CHEMISTRY 2019
Illustrations: Niklas Elmehed


DURACELL
PLUS
AA
AAA


DTP 603450
3.7V 1000mAh


EXIDE
MARINE & MULTIFIT
DUAL AGM

- ***Oxidation*** is defined as the ***loss of electrons*** from a compound (reactant) in a chemical reaction



$$M \longrightarrow M^{+} \; + \; 1e\text{-}$$

- The process of ***gaining electrons*** is called ***reduction***

- ***The compound helps oxidation is known as oxidizing agent. So oxidizing agent gain electrons***

- ***The compound helps reduction is known as reducing agent. So, they losses electrons***

QN

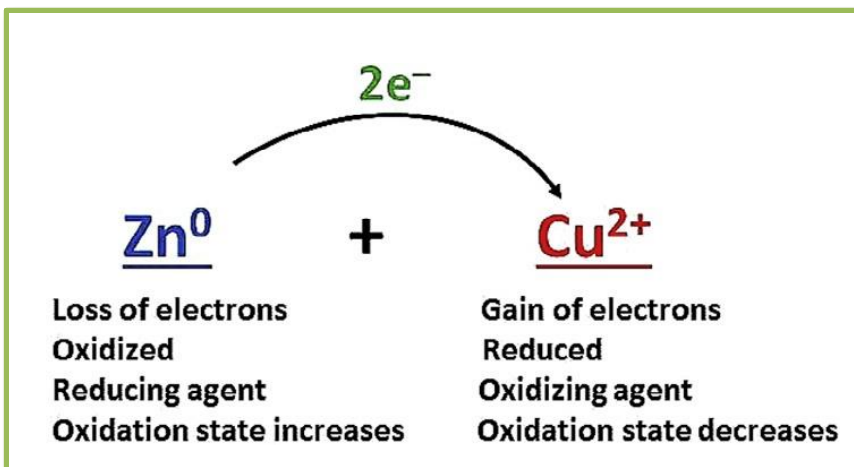
$$Zn + Cu^{2+} \longrightarrow Zn^{2+} + Cu$$

Oxidation reaction
Reduction reaction

Oxidizing agent
Reducing agent

$Zn \longrightarrow Zn^{2+}$ Oxidation reaction

$Cu^{2+} \longrightarrow Cu$ Reduction reaction



Solid-liquid interface

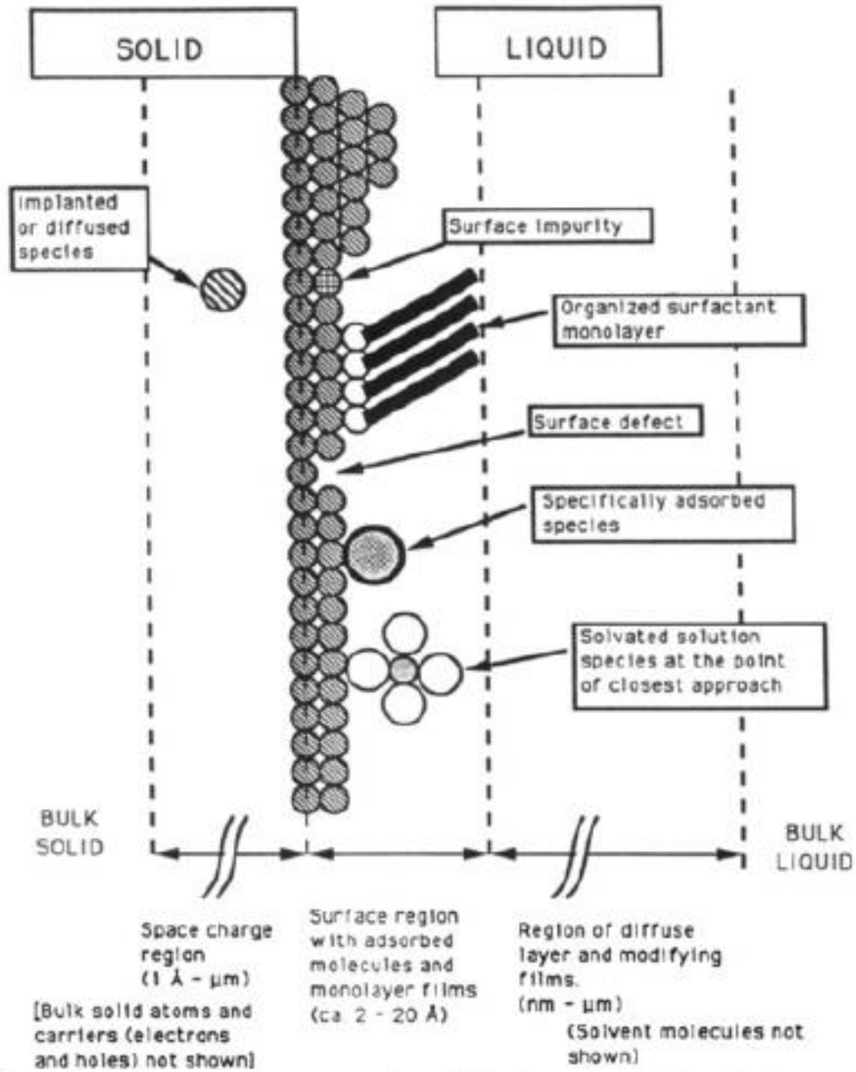


Figure 1. Schematic representation of the liquid–solid interfacial region showing possible species and features.[346]

The diagram shown is a representation of solid-liquid interface.

The interface is located in between bulk solid and bulk liquid.

It is divided into three regions:

1. **Surface charge region of the solid**:

2. Surface region of the solid with **adsorbed ions** and molecules.

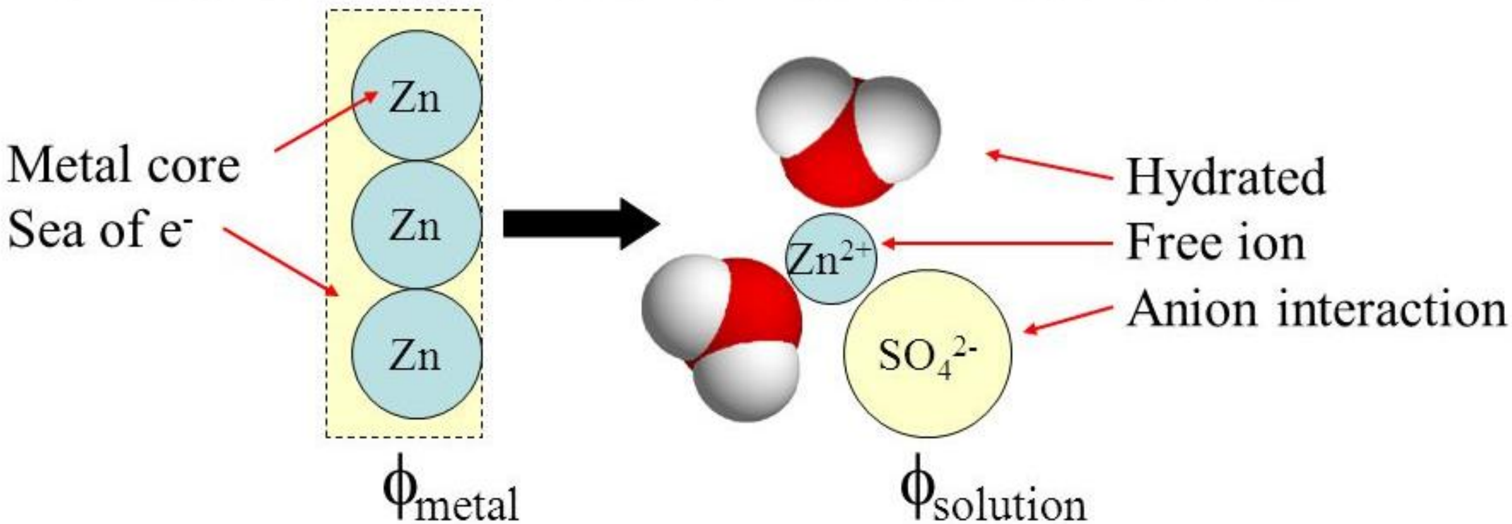
3. Region of **diffuse layer**:
A diffuse layer contains higher concentrations ions than the ions present in bulk liquid.

Interfacial Potential Difference

The charge separation at the interface makes the interface electrified.
It is called *electrified interface*.

Interfacial potential difference $\Delta\phi$

- arises from changes in chemical+electrical environments



$$\boxed{\Delta\phi = \phi_{metal} - \phi_{solution} = E \ (V)}$$

The interfacial potential difference is also called the electrode potential.

What is the origin of electrode potential?

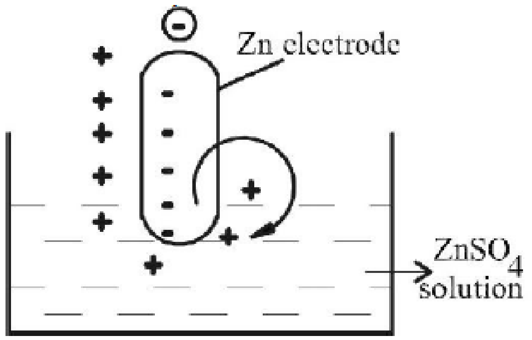
- When a metal (M) is placed in a solution of its own salt (M^{n+}) one of the *two processes are possible*

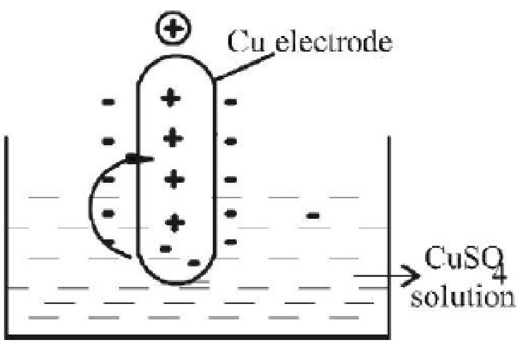
  (i) Metal atoms go into solution in the form of ions.

  $$\mathbf{M \rightarrow M^{n+} + ne^-} \quad \text{(Oxidation)}$$

  (ii) Metal ions from solution may deposit on the metal.

  $$\mathbf{M^{n+} + ne- \rightarrow M} \quad \text{(reduction)}$$

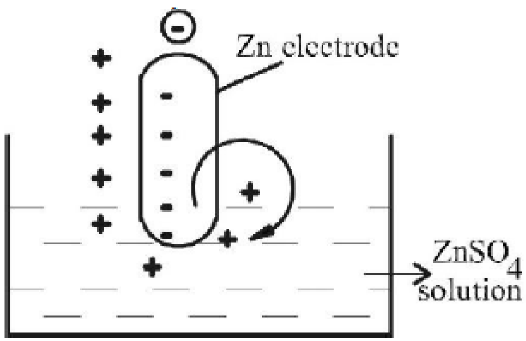


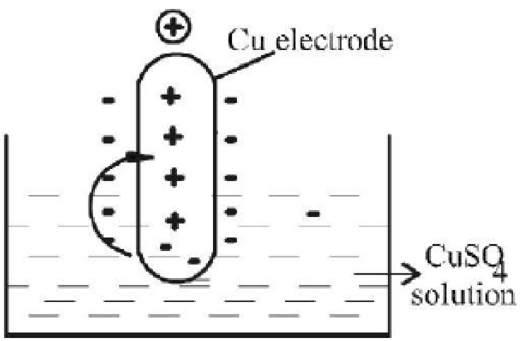


$Cu^{2+} + 2e- \rightarrow Cu$

The layer of charge developed at the electrode-electrolyte interface is called *Helmholtz electrical double layer*, which results a potential difference between the metal plate and ions in the solution.

Oxidation and Reduction Potential





At certain point the system reaches equilibrium.

At equilibrium, the potential difference becomes a constant value which is known as the electrode potential of the metal.

Thus the *tendency of the electrode to lose electrons* is called *oxidation potential* and tendency of an electrode to gain electrons is called reduction potential.

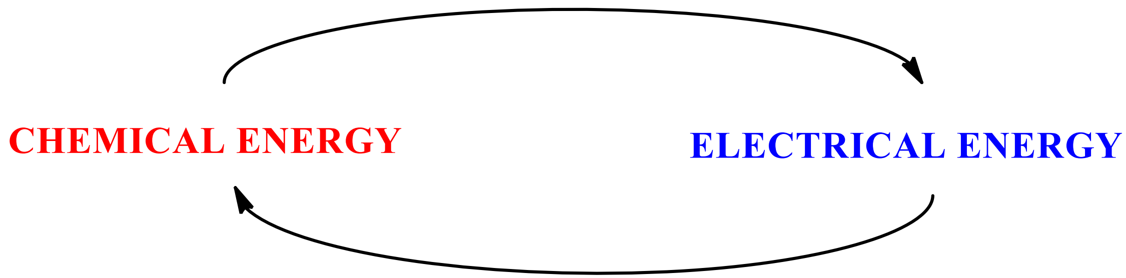
Single electrode (E) and standard electrode potential (E^o)

Single electrode potential (E) :
Single electrode potential is also referred as *half-cell potential*.
It is the tendency of a metallic electrode to lose or gain electrons when it is in contact with a solution of its own salt.

Standard electrode potential (E^o) :
It is the tendency of a metallic electrode to lose or gain electrons when it is in contact with a solution of its own salt of 1M concentration at 25^oC.
If it is a gas, 1 atm pressure has to be maintained.

Electrochemical Cell

- An **electrochemical cell** is a device capable of either generating electrical energy from chemical reactions or using electrical energy to cause chemical reactions



Two types of Electrochemical Cells:

1. **Galvanic cell:** An electrochemical cell, where ***spontaneous chemical change*** is used to produce electric current (**$\Delta G < 0$**).
2. **Electrolytic cell:** An electrochemical cell, where electric current is used to carry out a ***non-spontaneous chemical reaction***.

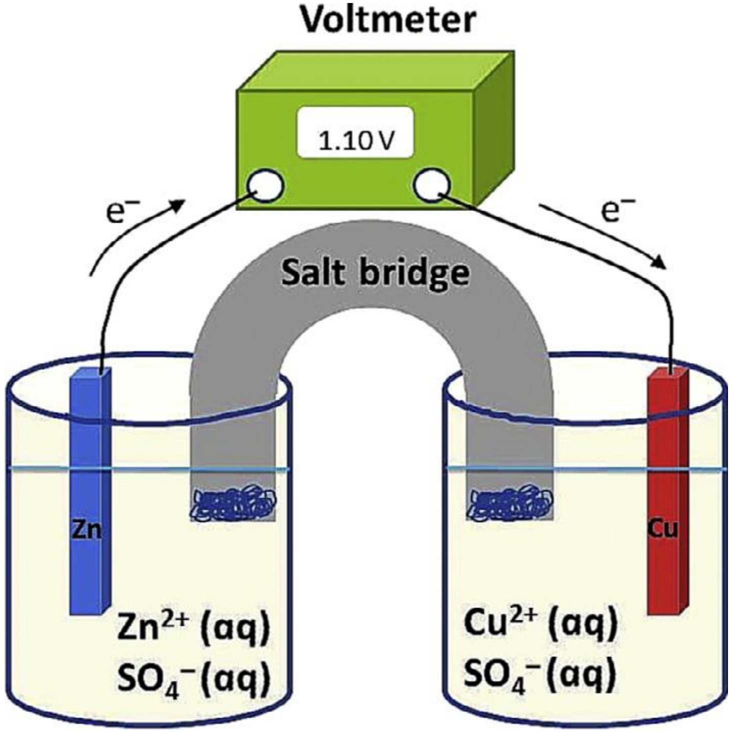
- The electrochemical cells which **generate an electric current** are called voltaic cell or **Galvanic cell**

Galvanic Cell

What is the need of a **salt bridge**?

Why two separate container/compartment?

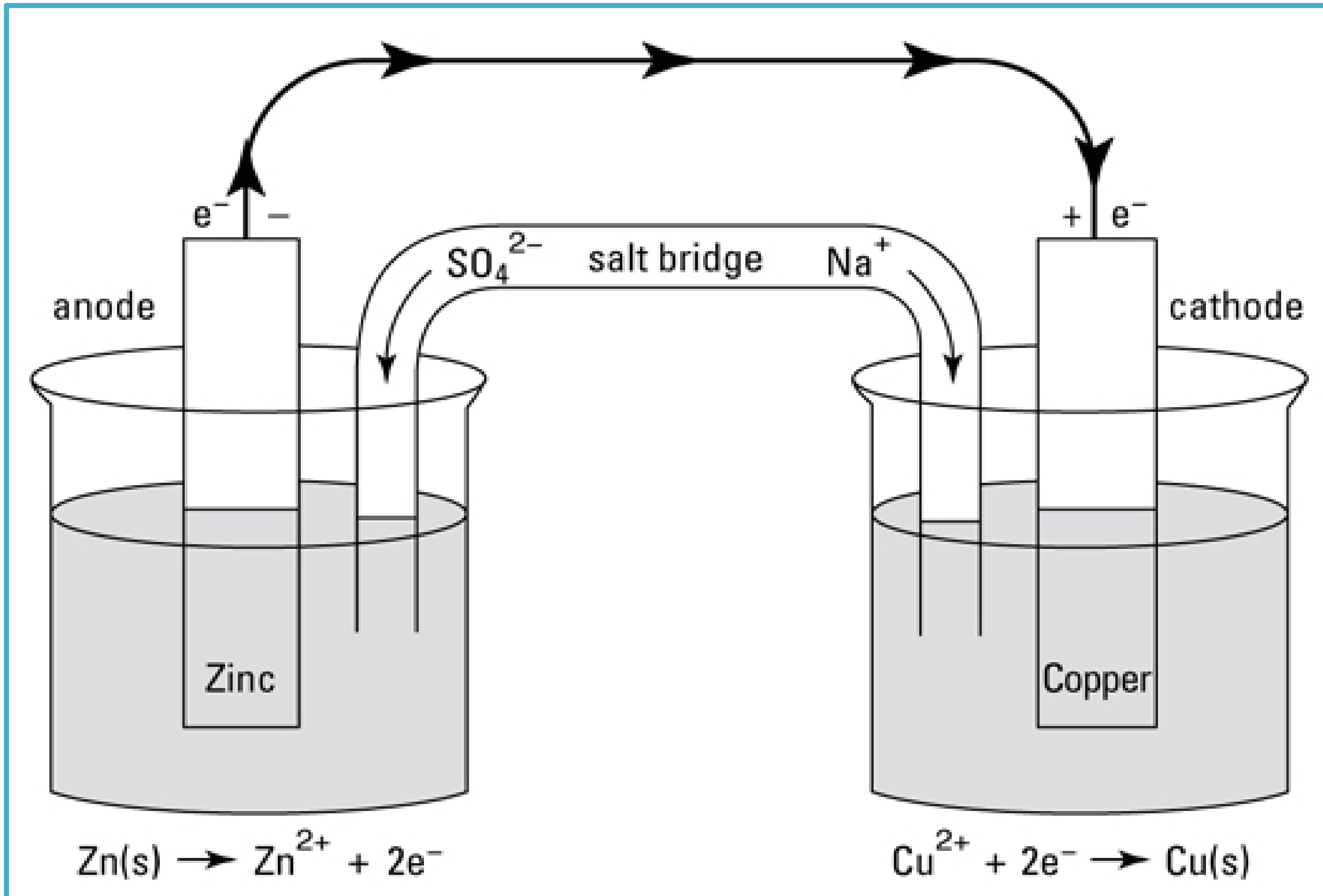
Which one is cathode, and which one anode? Why?



Conditions to make a battery?

Working of Galvanic Cell

- Electrons are being transferred in the Zn/Cu^{2+} reaction, but the system does not generate electrical energy because the oxidizing agent (Cu^{2+}) and the reducing agent (Zn) are in the same beaker.
- If, however, the half-reactions are physically separated and connected by an external circuit, the electrons are transferred by traveling through the circuit and an electric current is produced.
- This separation of half-reactions is the essential idea behind a voltaic cell.
- The components of each half-reaction are placed in a separate container, or half-cell, which consists of one electrode dipping into an electrolyte solution.
- The two half-cells are joined by the circuit, which consists of a wire and a salt bridge.


e⁻
−
+
e⁻
SO4 2−
salt bridge
Na+
anode
cathode
Zinc
Copper
Zn(s) → Zn2+ + 2e−
Cu2+ + 2e− → Cu(s)

Working of Galvanic Cell

Anode and Cathode

- **Oxidation** of an ion or molecule happed on the surface of an electrode, where it releases electrons in to the electrode, known as ***anode***.

- These electrons travel through external circuit to another electrode and being accepted by another ion or molecule, i.e. getting reduced.

- The electrode where **reduction** happens, known as ***cathode***.

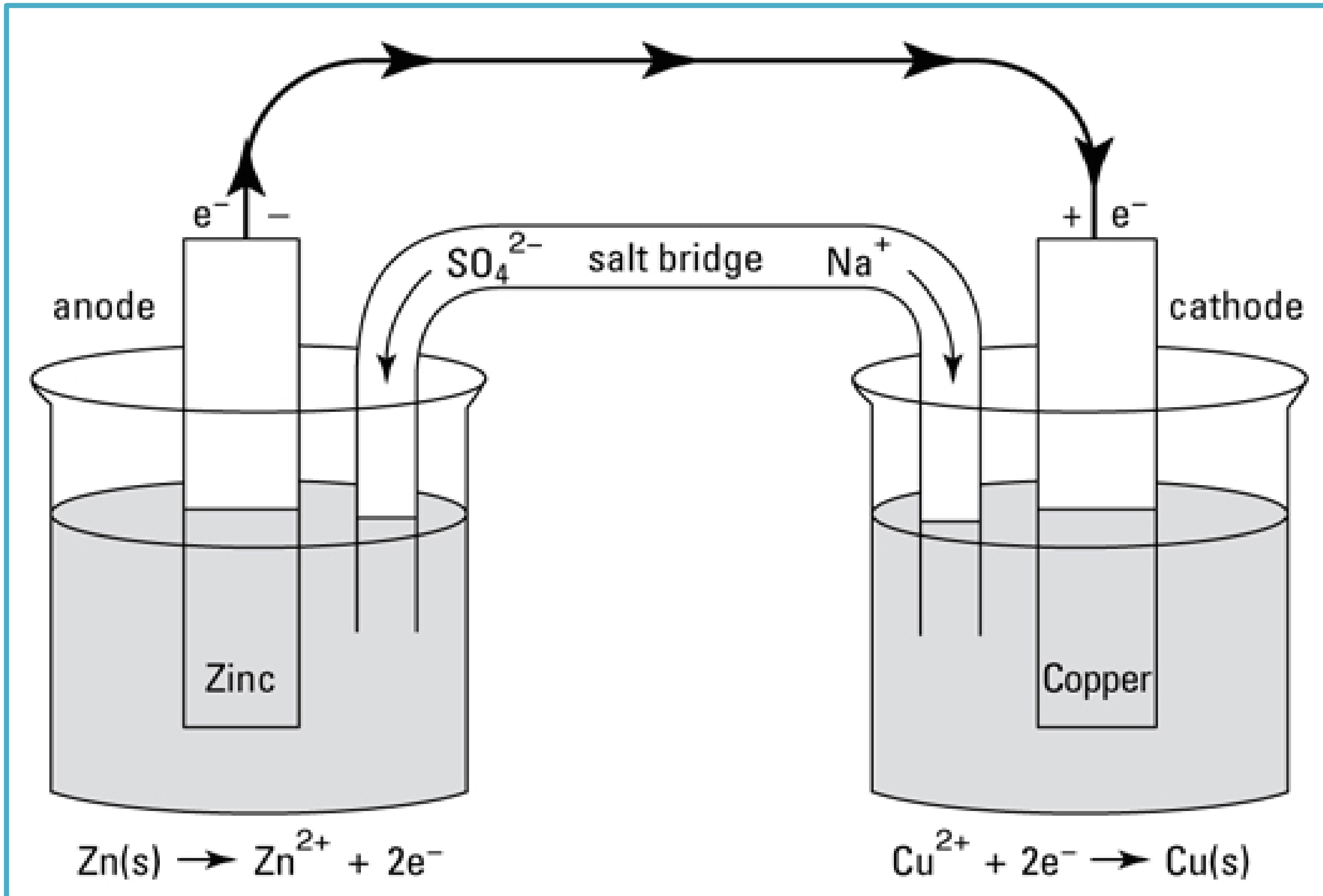
- The solution conducts electricity through the movement of ions, known as ***ionic conduction***. Cations travel towards cathode and anions travel towards anode (**to balance the charge!!**).

The purpose of the salt bridge

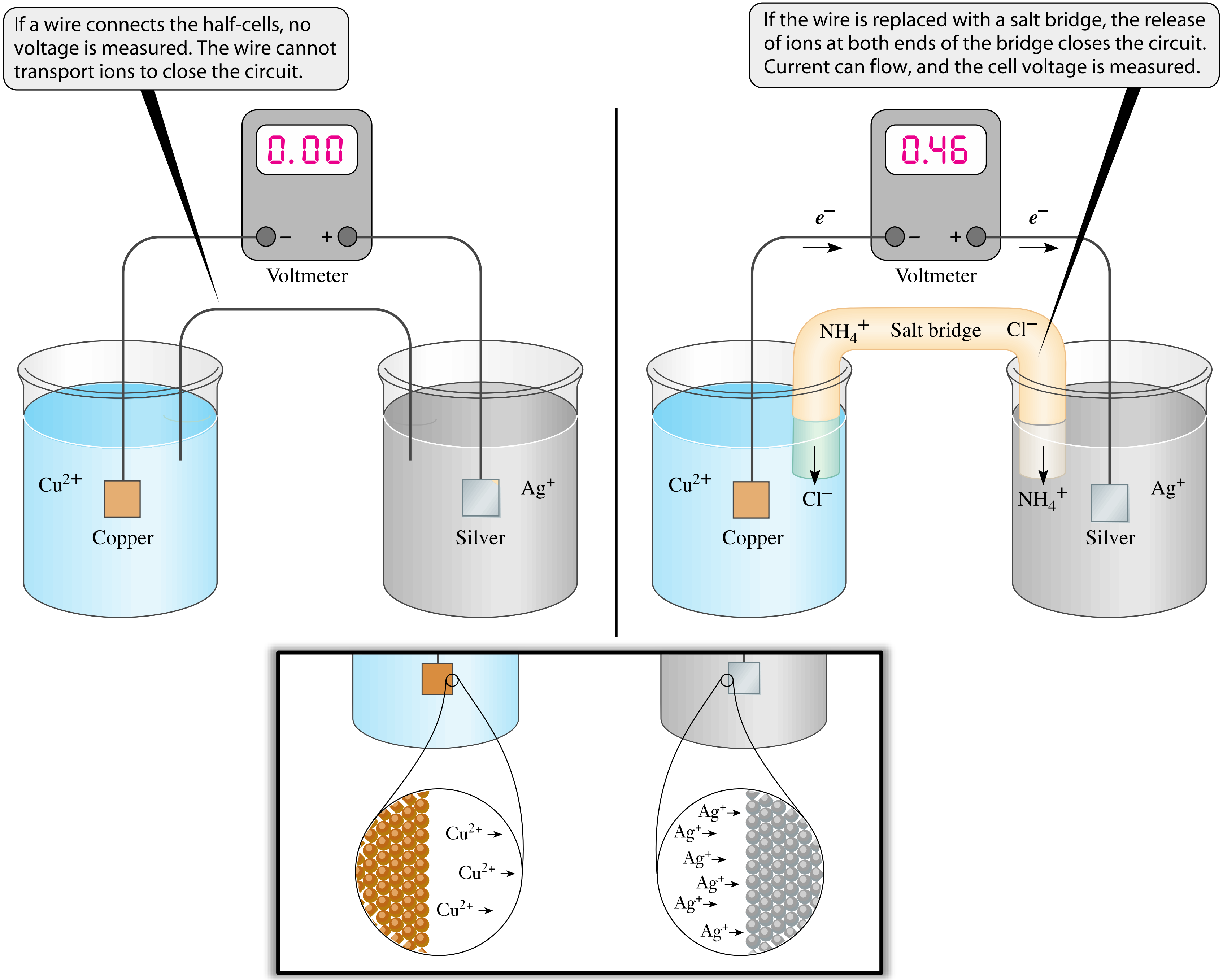
- The **oxidation half-cell** originally contains a neutral solution of Zn^{2+} and SO_4^{2-} ions, but as Zn atoms in the bar lose electrons, the solution would develop a **net positive charge** from the Zn^{2+} ions entering.

- Similarly, in the **reduction half-cell**, the neutral solution of Cu^{2+} and SO_4^{2-} ions would develop a **net negative charge** as Cu^{2+} ions leave the solution to form 'Cu' atoms.

- If the half-cells do not remain neutral, the resulting charge imbalance would stop cell operation. To avoid this situation and enable the cell to operate, the two half cells are joined by a **salt bridge,** which acts as a "**liquid wire**," **allowing ions to flow through both compartments and complete the circuit**



e⁻
−
+
e⁻
anode
cathode
SO_4^{2-}
salt bridge
Na^+
Zinc
Copper
$Zn(s) \rightarrow Zn^{2+} + 2e^-$
$Cu^{2+} + 2e^- \rightarrow Cu(s)$



If a wire connects the half-cells, no voltage is measured. The wire cannot transport ions to close the circuit.
0.00
Voltmeter
Cu^{2+}
Copper
Silver
Ag^+
If the wire is replaced with a salt bridge, the release of ions at both ends of the bridge closes the circuit. Current can flow, and the cell voltage is measured.
0.46
e^-
e^-
Voltmeter
NH_4^+
Salt bridge
Cl^-
Cu^{2+}
Cl^-
NH_4^+
Ag^+
Copper
Silver
Cu^{2+}
Cu^{2+}
Cu^{2+}
Ag^+
Ag^+
Ag^+
Ag^+
Ag^+
Ag^+

For this cell, the spontaneous reaction is,

$$Zn\,(s) + Cu^{2+}\,(aq) \longrightarrow Zn^{2+}\,(aq) + Cu\,(s)$$

Anode reaction: $Zn\,(s) \longrightarrow Zn^{2+}\,(aq) + 2\,e^-$

Cathode reaction: $Cu^{2+}\,(aq) + 2\,e^- \longrightarrow Cu\,(s)$

- The reaction at a particular electrode is called ***half reaction***.
- The reduced state and oxidized state of substance taking part in the half reaction form a ***redox couple***.

Half-reaction	Redox couple
$Cu^{2+}\,(aq) + 2\,e^- \longrightarrow Cu\,(s)$	Cu^{2+}/Cu
$Zn^{2+}\,(aq) + 2\,e^- \longrightarrow Zn\,(s)$	Zn^{2+}/Zn

Cell diagram / Representation

- A ***vertical line*** (|) is used to present a *metal-solution interface*, and *junction of different phases*

- If the oxidized and reduced species are in same phase then a ***comma*** (,) is used

- The metal is written first for **anode** and at last for **cathode**

- A ***double vertical line*** (||) is used to show a *salt bridge* or *porous cup*

Salt bridge

$$\overbrace{\mathrm{Zn}(s)\,|\,\mathrm{Zn}^{2+}(aq)}^{\text{Anode half-cell}}\;\|\;\overbrace{\mathrm{Cu}^{2+}(aq)\,|\,\mathrm{Cu}(s)}^{\text{Cathode half-cell}}$$

Phase boundary　　Electrons flow this way　　Phase boundary

Cell Potential (E_{cell}) (emf)
Out-put of a Galvanic Cell

- The purpose of a voltaic cell is to convert the free energy change of a **spontaneous reaction** into the kinetic energy of electrons moving through an external circuit (**electrical energy**).

- This electrical energy is proportional to the *difference in electrical potential between the two electrodes*, which is called the **cell potential (E_{cell});** it is also called the **voltage** of the cell or the **electromotive force (emf).**

- For spontaneous cell operation: $E_{\text{cell}} > 0$ for a spontaneous process

- ***Greater the emf, more the energy released from electron flow through wire***.

Cell Potential:

- The energy available form a cell depends on how fast it can travel from anode to cathode.
- It depends on how fast electrons can be pushed from anode, and how fast it can be accepted at cathode.

The pushing-pulling effect of a cell is measured by its ***cell potential (E)***.

$$1 \text{ volt (V)} = \frac{1 \text{ joule (J)}}{1 \text{ coulomb (C)}}$$

$$\text{Energy (J)} = \text{charge (c)} \text{ ´ } \text{cell potential (V)}$$

Cell potential is a positive quantity.

A cell has 1.5 V emf means with passing of 1 C of charge from anode to cathode produces it produces (1 C $\times$ 1.5 V) = 1.5 J energy.

A charge of 1 C = 6.2×10^{18} electrons, as 1 electron has 1.602×10^{-19} C

Electrode Potential &
Standard Electrode Potential (E^0_{cell})

- The cell potential is combination of individual contribution from the electrodes. This individual contribution is called ***electrode potential***

- The measured potential of a voltaic cell is affected by changes in **concentration** as the reaction proceeds and by energy losses due to heating of the cell and the external circuit.

- Therefore, in order to compare the output of different cells, we obtain a **standard cell potential (E^0_{cell}),** the potential measured at a specified temperature (usually **298 K**) with no current flowing and *all components in their standard states:*

1 atm for gases, **1 *M*** for solutions, **298 K**, the **pure solid** for electrodes.

Calculation of Cell potential

- The **standard electrode potential ($E^0_{half\text{-}cell}$)** is the potential associated with a given half-reaction (electrode compartment) when all the components are in their standard states.

- By convention, *a standard electrode potential always refers to the half reaction written as a* ***reduction***

$$\mathrm{Zn^{2+}}(aq) + 2\mathrm{e^-} \longrightarrow \mathrm{Zn}(s) \qquad E^\circ_{\mathrm{zinc}}\ (E^\circ_{\mathrm{anode}}) \qquad \text{[reduction]}$$

$$\mathrm{Cu^{2+}}(aq) + 2\mathrm{e^-} \longrightarrow \mathrm{Cu}(s) \qquad E^\circ_{\mathrm{copper}}\ (E^\circ_{\mathrm{cathode}}) \qquad \text{[reduction]}$$

- The Over all cell potential is calculated as,

$$E^\circ_{\mathrm{cell}} = E^\circ_{\mathrm{copper}} - E^\circ_{\mathrm{zinc}}$$

$$\boxed{E^0_{cell} = E^0_{cat\,hode} - E^0_{Anode}}$$

Standard hydrogen electrode (SHE)

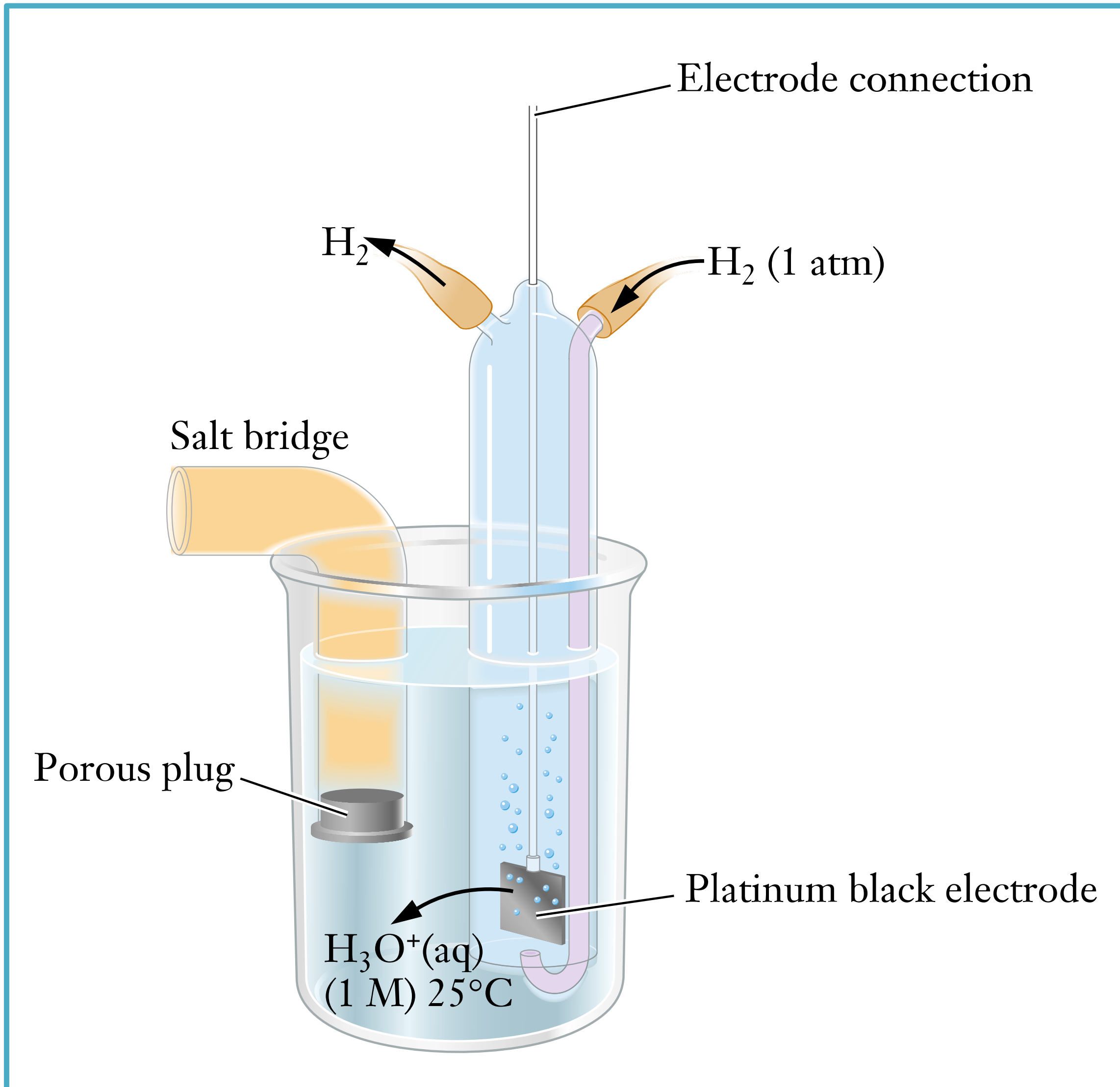
- H^+/H_2 redox couple at standard state (H^+ ion concentration = 1 mol/L, and H_2 pressure = 1 atm) is taken as **reference, i.e. E^o is 0**

$$2\,H^+(aq) + 2\,e^- \rightarrow H_2(g)$$

$$Pt(s) \mid H_2(g, 1\text{ atm}) \mid H^+(1\,M)$$

- To determine the half-cell potential of any other electrode/electrolyte system, it can be connected to the **SHE** and the cell potential measured

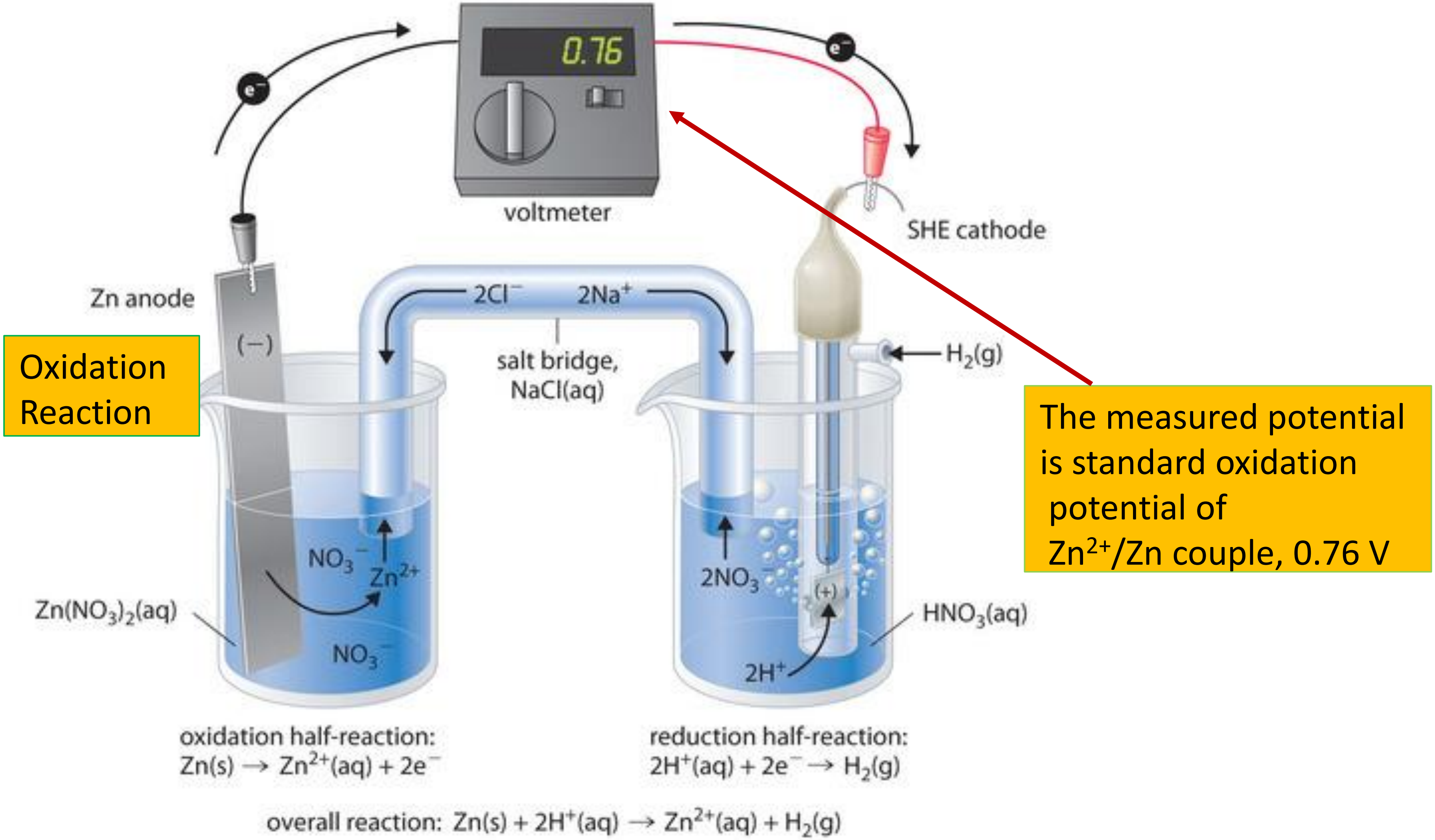
Standard hydrogen electrode (SHE)



When the measured cell potential is ***positive***, the SHE is the anode, and we know that **H_2 is oxidized to H^+ (aq)**

When the cell potential is ***negative***, the SHE is the cathode, and **H^+ (aq) is reduced to H_2**

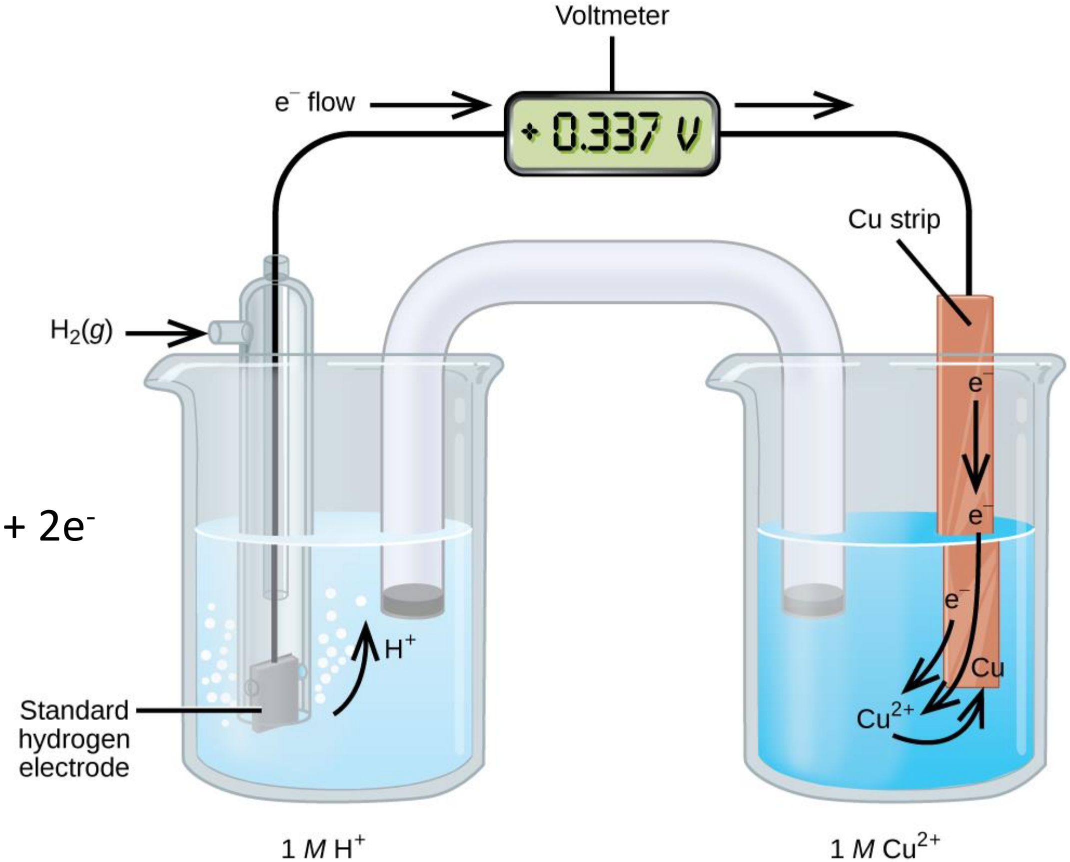
Half-Reaction	Standard Reduction Potential (V)
$Zn^{2+} + 2\,e^- \rightarrow Zn$	-0.763
$Fe^{2+} + 2\,e^- \rightarrow Fe$	-0.44
$2\,H^+ + 2\,e^- \rightarrow H_2$	0.000
$Cu^{2+} + 2\,e^- \rightarrow Cu$	$+0.337$
$Fe^{3+} + e^- \rightarrow Fe^{2+}$	$+0.771$
$Ag^+ + e^- \rightarrow Ag$	$+0.7794$


0.76
voltmeter
SHE cathode
Zn anode
(−)
Oxidation Reaction
2Cl⁻
2Na⁺
salt bridge, NaCl(aq)
H$_2$(g)
The measured potential is standard oxidation potential of Zn^{2+}/Zn couple, 0.76 V
NO_3^-
Zn^{2+}
$2NO_3^-$
(+)
$Zn(NO_3)_2$(aq)
HNO_3(aq)
NO_3^-
$2H^+$
oxidation half-reaction:
$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$
reduction half-reaction:
$2H^+(aq) + 2e^- \rightarrow H_2(g)$
overall reaction: $Zn(s) + 2H^+(aq) \rightarrow Zn^{2+}(aq) + H_2(g)$

MEASURING STANDARD ELECTRODE POTENTIAL OF CU^{2+}/Cu REDOX COUPLE

Anode

$H_2(g) \rightarrow 2H^+ (aq) + 2e^-$



Cathode

$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s$

Is the measured voltage is reduction or oxidation potential of Cu^{2+}/Cu couple?

The Electrochemical Series (STD Reduction Potential)

equilibrium	E° (volts)
$Li^+_{(aq)} + e^- \rightleftharpoons Li_{(s)}$	-3.03
$K^+_{(aq)} + e^- \rightleftharpoons K_{(s)}$	-2.92
$Ca^{2+}_{(aq)} + 2e^- \rightleftharpoons Ca_{(s)}$	-2.87
$Na^+_{(aq)} + e^- \rightleftharpoons Na_{(s)}$	-2.71
$Mg^{2+}_{(aq)} + 2e^- \rightleftharpoons Mg_{(s)}$	-2.37
$Al^{3+}_{(aq)} + 3e^- \rightleftharpoons Al_{(s)}$	-1.66
$Zn^{2+}_{(aq)} + 2e^- \rightleftharpoons Zn_{(s)}$	-0.76
$Fe^{2+}_{(aq)} + 2e^- \rightleftharpoons Fe_{(s)}$	-0.44
$Pb^{2+}_{(aq)} + 2e^- \rightleftharpoons Pb_{(s)}$	-0.13
$2H^+_{(aq)} + 2e^- \rightleftharpoons H_{2(g)}$	0
$Cu^{2+}_{(aq)} + 2e^- \rightleftharpoons Cu_{(s)}$	+0.34
$Ag^+_{(aq)} + e^- \rightleftharpoons Ag_{(s)}$	+0.80
$Au^{3+}_{(aq)} + 3e^- \rightleftharpoons Au_{(s)}$	+1.50

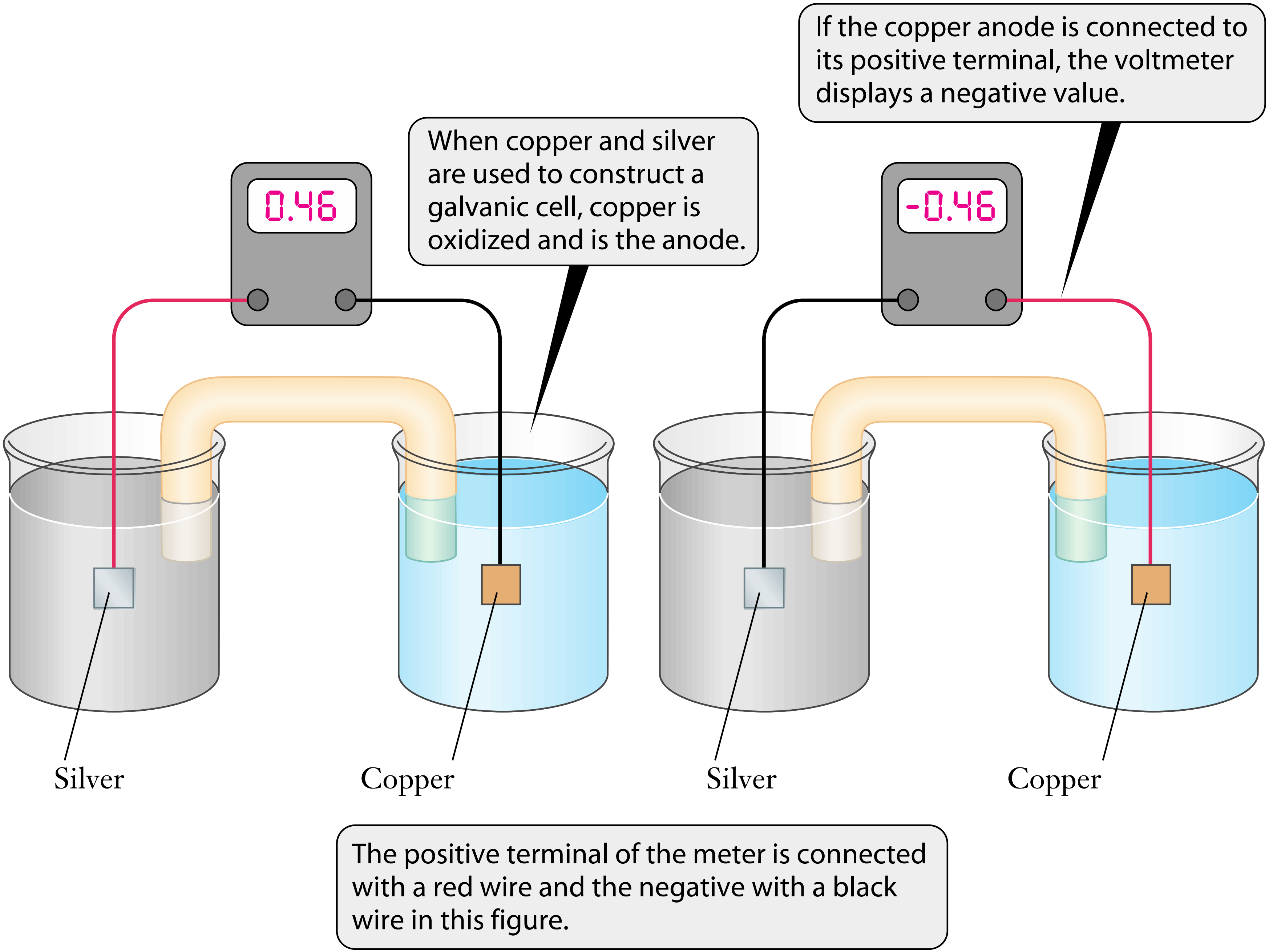
The electrochemical series is built up by arranging various redox equilibria in order of their standard reduction potentials (M^{n+}/M).
The most negative E° values are placed at the top of the electrochemical series, and the most positive at the bottom.

Application of electrochemical Series

1. To choose anode and cathode metal for batteries.

2. To understand the reactivity of metals.

3. To understand corrosion properties and cathodic protection of steel.
 (Explained latter)


If the copper anode is connected to its positive terminal, the voltmeter displays a negative value.
When copper and silver are used to construct a galvanic cell, copper is oxidized and is the anode.
0.46
-0.46
Silver
Copper
Silver
Copper
The positive terminal of the meter is connected with a red wire and the negative with a black wire in this figure.

QN1

PROBLEM A voltaic cell houses the reaction between aqueous bromine and zinc metal:

$$Br_2(aq) + Zn(s) \longrightarrow Zn^{2+}(aq) + 2Br^-(aq) \qquad E^\circ_{cell} = 1.83\ V$$

Calculate $E^\circ_{bromine}$, given $E^\circ_{zinc} = -0.76$ V.

SOLUTION Dividing the reaction into half-reactions:

$$Br_2(aq) + 2e^- \longrightarrow 2Br^-(aq) \qquad E^\circ_{unknown} = E^\circ_{bromine} = ?\ V$$

$$Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^- \qquad E^\circ_{zinc} = -0.76\ V$$

Here. Br_2 is getting reduced to Br-. i.e, the reduction reaction. Therefor, it will act as CATHODE

For zinc, oxidation happened. Therefor, 'Zn' will act as ANODE

Calculating $E^\circ_{bromine}$:

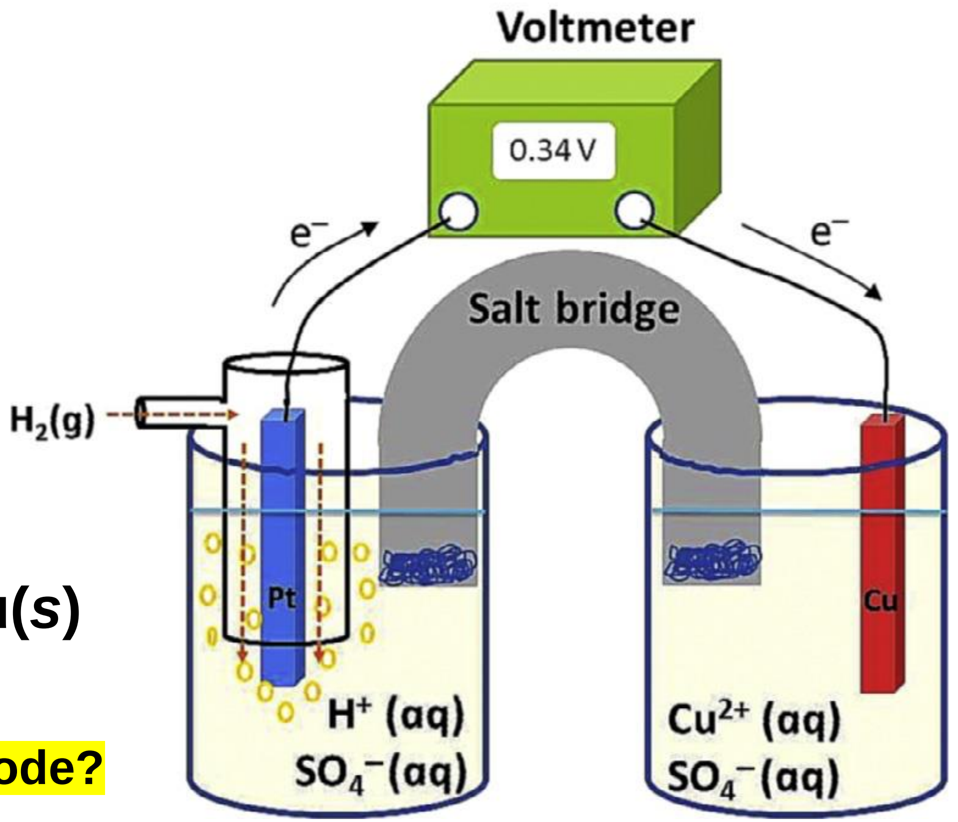
$$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode} = E^\circ_{bromine} - E^\circ_{zinc}$$

$$E^\circ_{bromine} = E^\circ_{cell} + E^\circ_{zinc} = 1.83\ V + (-0.76\ V) = \boxed{1.07\ V}$$

QN2

Here is SHE and Cu' cell.

And cell representation also given.



$$Pt \,|\, H_2(g) \,|\, H^+(aq) \,||\, Cu^{2+}(aq) \,|\, Cu(s)$$

Which one will act as Anode and Cathode?

Write the half-cell reactions

Cathode Reaction: $Cu^{2+}(aq) + 2e^- \longrightarrow Cu^{2+}(s) \qquad E^0 = +0.34\ V$

Anode Reaction: $H_2(g) \longrightarrow 2H^+(aq) + 2e^- \qquad E^0 = 0.00\ V$

QN3

$Zn(s) \mid Zn^{2+}(aq) \parallel H^+(aq) \mid H_2(g) \mid Pt$ $E^o = -0.76\ V$

Cell representation is given. Write the anode and cathode reactions ?

QN4

A galvanic cell with a measured standard cell potential of **0.27 V** is constructed using two beakers connected by a salt bridge. One beaker contains a strip of **gallium metal** immersed in a 1 M solution of $GaCl_3$, and the other contains a piece of **nickel** immersed in a 1 M solution of $NiCl_2$.

The half-reactions that occur when the compartments are connected are as follows:

cathode: $Ni^{2+}(aq) + 2e^- \rightarrow Ni(s)$

anode: $Ga(s) \rightarrow Ga^{3+}(aq) + 3e^-$

If the potential for the oxidation of **Ga to Ga^{3+} is 0.55 V** under standard conditions, **what is the potential for the oxidation of Ni to Ni^{2+}?**

Ans.

Step 1. The standard oxidation potential Ga/Ga^{3+} = 0.55 V

Convert this to reduction potential,

Ga^{3+}/Ga = - 0.55 V (opposite sign)

Step 2 Calculate the standard reduction potential of Ni^{2+}/Ni couple.

$$E^0_{cell} = E^o_{cathode} - E^o_{anode}$$

$E^0_{cell} = 0.22$ V

$E^o_{cathode} = E^o_{Ni2+/Ni} = ?$

$E^o_{anode} = E^o_{Ga3+/Ga} = -0.55$ V

$$E^o_{Ni2+/Ni} = 0.22 - 0.55 = -0.28 \text{ V}$$

This is the reduction potential of Ni^{2+} to Ni. But, we have to calculate 'oxidation potential.

Step 3.

Reverse the sign to get oxidation potential

$$E^o_{Ni2+/Ni} = -0.28 \text{ V}$$

$$E^o_{Ni/Ni2+} = +0.28 \text{ V}$$

QN5

For the given half-cells which one will be anode and which one will be cathode?

Zn^{2+}/Zn and Li^{+}/Li

(Given: $E^0_{Zn2+/Zn} = - 0.763$ and $E^0_{Li+/Li} = - 3.04$)

Ans.

- Lithium metal half-cell pair lies left with respect to the zinc metal pair in electrochemical series.
- Therefore lithium metal in Li^{+}/Li pair has higher tendency to lose electrons.
- In other words it has more negative electrical potential. This half-cell pair has to form the anode reaction.
- Zn^{2+}/Zn will form cathode reaction.

Why iron rusts in open atmosphere?
Explain with the help of electrochemical series

$Fe^{2+} + 2e^- \rightarrow Fe(s)$ -0.44 V

$O_2 + 4e^- \rightarrow 2\,O^{2-}$ +1.23 V

The standard reduction potential of iron-couple is < oxygen couple

Therefore, Fe acts as anode and gets oxidized by O_2

equilibrium	E° (volts)
$Li^+_{(aq)} + e^- \rightleftharpoons Li_{(s)}$	-3.03
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$Ca^{2+}_{(aq)} + 2e^- \rightleftharpoons Ca_{(s)}$	-2.87
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$Fe^{2+}_{(aq)} + 2e^- \rightleftharpoons Fe_{(s)}$	-0.44
$Pb^{2+}_{(aq)} + 2e^- \rightleftharpoons Pb_{(s)}$	-0.13
$2H^+_{(aq)} + 2e^- \rightleftharpoons H_{2(g)}$	0
$Cu^{2+}_{(aq)} + 2e^- \rightleftharpoons Cu_{(s)}$	+0.34
$Ag^+_{(aq)} + e^- \rightleftharpoons Ag_{(s)}$	+0.80
$Au^{3+}_{(aq)} + 3e^- \rightleftharpoons Au_{(s)}$	+1.50

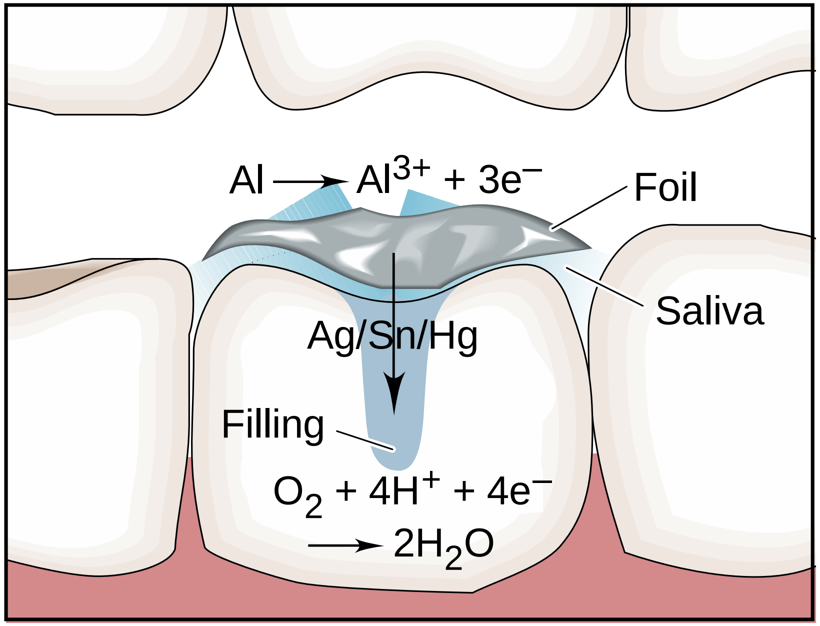
Why gold does not tarnish whereas silver tarnishes in open atmosphere?

The standard reduction potential of ***gold-couple is more positive (1.5 V)*** as compared to the oxygen-couple
$O_2 + 4e^- \rightarrow 2\ O^{2-}$ +1.23 V

Therefore gold does not acts as anode, it does not tarnish

Whereas for silver, the opposite tendency occurs

Standard reduction potential of silver-couple is < oxygen couple.
Here, silver acts as anode and gets tarnished in open air.



The Pain of a Dental Voltaic Cell

Have you ever felt a jolt of pain when biting down with a filled tooth on a scrap of foil left on a piece of food? Here's the reason. The aluminum foil acts as an active anode (E° of Al $= -1.66$ V), saliva as the electrolyte, and the filling (usually a silver/tin/mercury alloy) as an inactive cathode. O_2 is reduced to water, and the short circuit between the foil in contact with the filling creates a current that is sensed by the nerve of the tooth.

Dental filling is an Ag/Sn/Hg alloy.

When a person with dental filling bit a piece of food that contains small aluminum foil will experience an electric shock.

Non-Standard State Cell Potential (E_{cell})
and
Nernst Equation

Free Energy & E_{cell}
Thermodynamic Aspects

- A spontaneous reaction has a *negative* free energy change ($\Delta G < 0$), and you've just seen that a spontaneous electrochemical reaction has a *positive* cell potential ($E_{cell} > 0$).

- Note that *the signs of ΔG and E_{cell} are opposite for a spontaneous reaction.* These two indications of spontaneity are proportional to each other:

$$\Delta G \propto -E_{\text{cell}}$$

- When all of the components are in their standard states, we have

$$\Delta G^\circ = -nFE^\circ_{\text{cell}}$$

Where, n is the number of moles of electrons transferred and **F** is **Faraday Constant**, F = 96500 *C*

Electrochemistry and Thermodynamics:

ΔG, free energy, is negative and large for a spontaneous change. If a cell reaction has a large negative ΔG, then the cell potential should be high. When the ΔG is 0 then the cell potential will be 0. Therefore,

$$\Delta G \propto -E_{\text{cell}}$$

$$\Delta G^{\circ} = -nFE^{\circ}_{\text{cell}}$$

F is called ***Faraday's Constant***, is the magnitude of charge per mole of electron.

F = charge of one elecron ´ number of electrons per mole

= 1.602 ´ 10^{-19} C ´ 6.022 ´ 10^{23} /mol e^-

= 9.647 ´ 10^4 C/mol e^- **= 96500 C**

'n' is the number of moles of electrons transferred for one mole or cell reaction.

Nernst Equation

- Gibbs energy varies with ratio of reaction and products, therefore, E_{cell} *must vary with the concentration*.
- *Nernst equation is used to calculate the* ***E_{cell} under non-standard state*** *where the solute concentration is not 1M"*

$$\Delta G = \Delta G^o + RT \ln Q$$

$$\Delta G = -nFE_{cell} \quad \text{and} \quad \Delta G^o = -nFE^o_{cell}$$

$$E_{cell} = E^o_{cell} - \frac{RT}{nF} \ln\ Q$$

$$E_{cell} = E^o_{cell} - \frac{0.0257}{n} \ln\ Q$$

$$\boxed{E_{cell} = E^o_{cell} - \frac{0.0592}{n} \log_{10}\ Q}$$

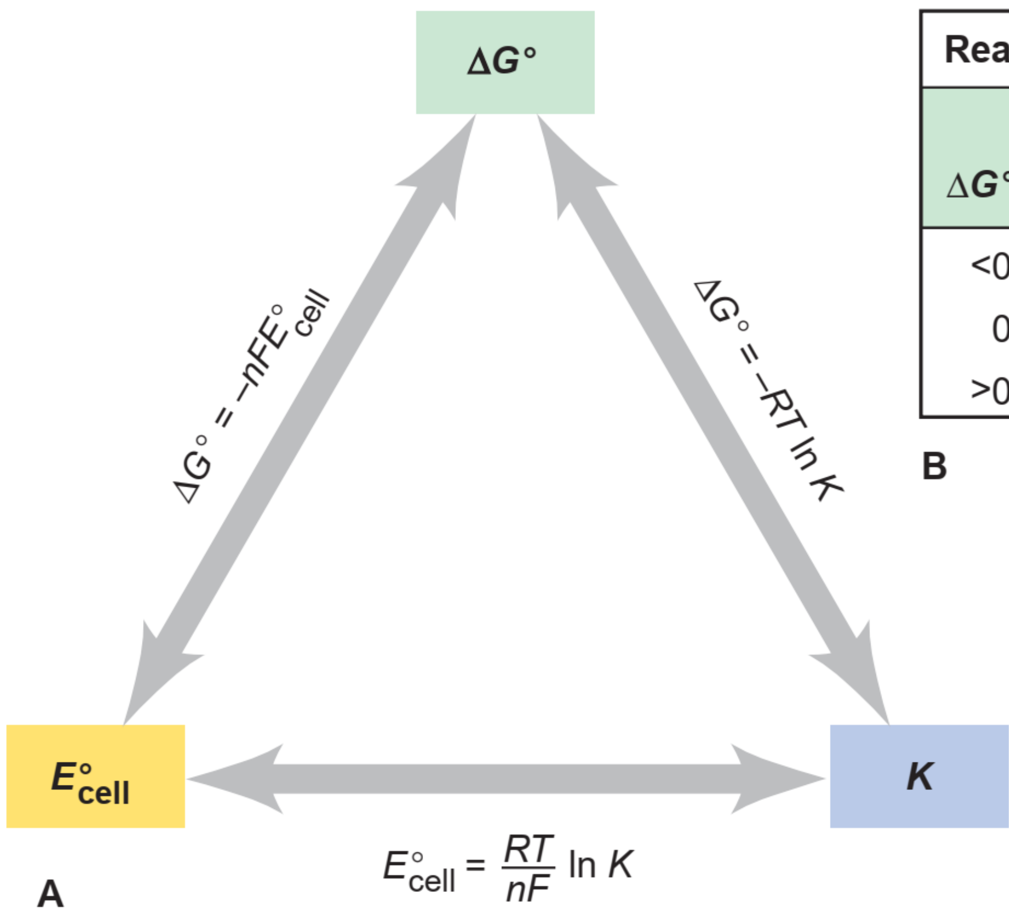
Nernst equation

$$E_{cell} = E^{o}_{cell} - \frac{0.0592}{n}\log_{10}\ Q$$

$$\mathbf{Q} = \frac{[\boldsymbol{Product}]}{[\boldsymbol{Reactant}]}$$

- When $Q < 1$ and thus [reactant] $>$ [product], $\ln Q < 0$, so $E_{\text{cell}} > E^{\circ}_{\text{cell}}$.
- When $Q = 1$ and thus [reactant] $=$ [product], $\ln Q = 0$, so $E_{\text{cell}} = E^{\circ}_{\text{cell}}$.
- When $Q > 1$ and thus [reactant] $<$ [product], $\ln Q > 0$, so $E_{\text{cell}} < E^{\circ}_{\text{cell}}$.

Relationship with Thermodynamic Parameters



A

Reaction Parameters at the Standard State			
ΔG°	K	E°_{cell}	Reaction at standard-state conditions
<0	>1	>0	Spontaneous
0	1	0	At equilibrium
>0	<1	<0	Nonspontaneous

B

Figure 21.10 The interrelationship of ΔG°, E°_{cell}, and K. A, Any one of these three central thermodynamic parameters can be used to find the other two. **B,** The signs of ΔG° and E°_{cell} determine the reaction direction at standard-state conditions.

$$E_{cell} = E^{o}_{cell} - \frac{0.0592}{n}\log_{10}\ Q$$

If E_{cell} = 0, then

$$E^{\circ} = \frac{0.0592\ \mathrm{V}}{n}\log K$$

If we choose an electrochemical reaction that transfers only one electron at a potential of 1.0 V, and we determine the value of K, we obtain:

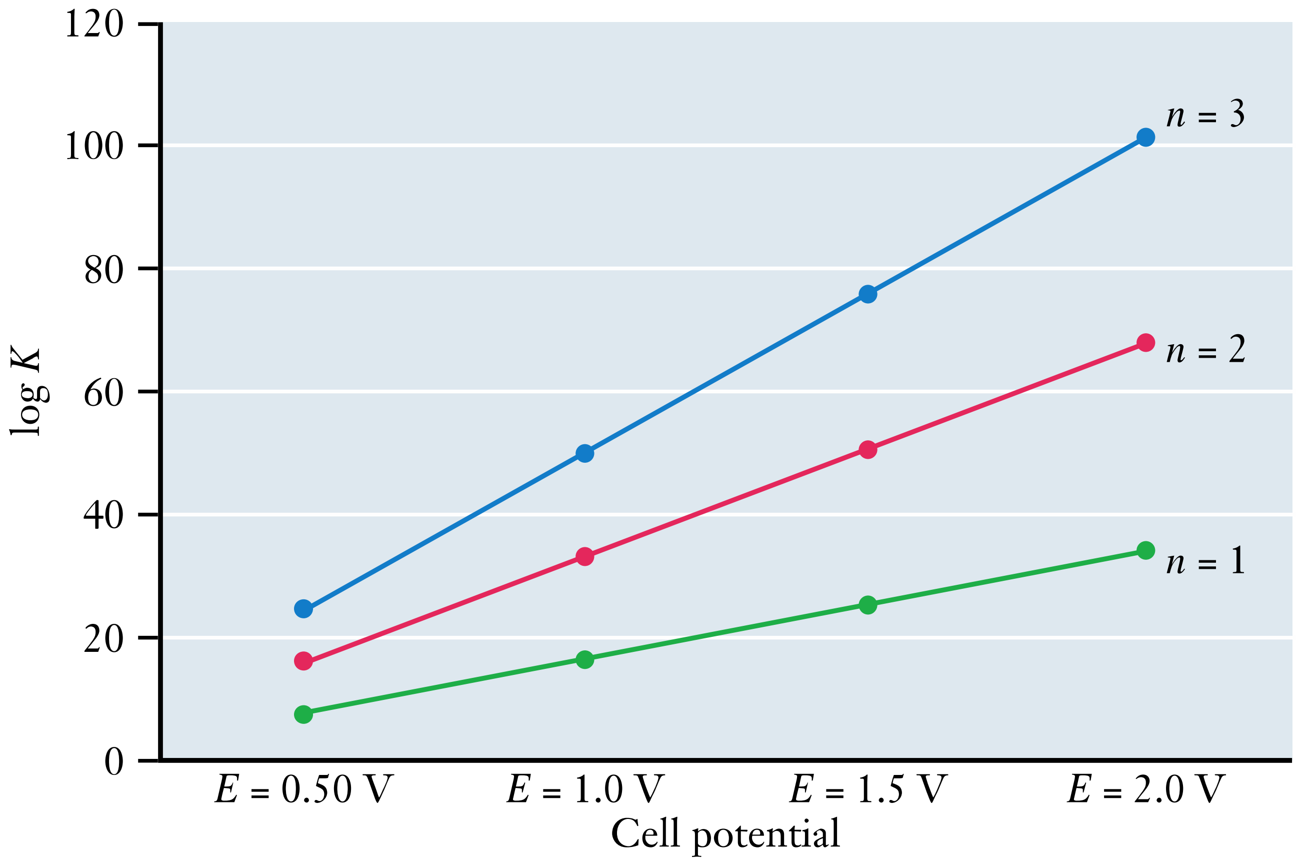
$$K = 10^{\frac{(1)(1.0\ \mathrm{V})}{0.0592\ \mathrm{V}}} = 7.8 \times 10^{16}$$

Large values of K indicate that equilibrium strongly favors products

$$E_{cell} = E^{o}_{cell} - \frac{0.0592}{n}\log_{10}\ Q$$

$$E^{\circ} = \frac{0.0592\ \mathrm{V}}{n}\log K$$

$$\log \mathrm{K} = \mathrm{E}^{0}{}_{\mathrm{cell}} \times \frac{n}{0.0592}$$



Slope ∝ n

How concentration affects the E_{cell} and the tendency of product formation?

$$E_{cell} = E^o_{cell} - \frac{RT}{nF}\ln\ Q$$

Q = [Products]/[Reactants]

Nernst equation that for a reaction in which n = 1,

25.7 mV

$$E_{cell} = E^o_{cell} - \frac{RT}{\cancel{nF}}\ln\ Q$$

RT/F = 0.0257 V = 25.7 mV

if *Q* is decreased by a factor of 10, then the cell potential becomes more positive by 25.7 mV x ln 10 = 59.2 mV. The reaction has a greater tendency to form products.

If *Q* is increased by a factor of 10, then the cell potential falls by 59.2 mV and the reaction has a lower tendency to form products.

Problem 1

$$Zn\,(s) \;+\; Cu^{2+}\,(aq) \rightleftharpoons Zn^{2+}\,(aq) \;+\; Cu\,(s)$$

For this cell reaction, the E° is 1.10 V, n = 2,

Then calculate equilibrium constant (K_c)

Ans.

$$E^{\circ}_{\mathrm{cell}} = \frac{RT}{nF}\ln K$$

$$\ln K_c = \frac{2 \times 1.10\ \mathrm{V}}{0.02569\ \mathrm{V}} = 85.6$$

$$\text{or, } K_c = 1.6 \times 10^{37}$$

Problem 2

What is the value of E_{cell} for the voltaic cell below

$$Pt_{(s)}|Fe^{2+}_{(0.1M)}, Fe^{3+}_{(0.2M)}||Ag^{+}_{(0.1M)}|Ag_{(s)}$$

$E^0(Ag^+/Ag)$ = 0.80 V $E^0(Fe^{3+}/Fe^{2+})$ = 0.771 V

Ans.

$$\boxed{E_{cell} = E^o_{cell} - \frac{0.0592}{n}\log_{10} Q}$$

Step 1. **Find the E^0_{cell}**

$$E^o_{cell} = E^o_{cathode} - E^o_{anode}$$

$$= E^o_{Ag/Ag} - E^o_{Fe^{3+}/Fe^{2+}}$$

$$= 0.800V - 0.771V = 0.029V$$

Step 2

Now to determine Ecell for the reaction

$$Fe^{2+}(0.1M) + Ag^{+}(1.0M) \rightarrow Fe^{3+}(0.20M) + Ag(s)$$

No of e- (n) = 1

Use the Nernst equation

$$\boxed{E_{cell} = E^{o}_{cell} - \frac{0.0592}{n}\log_{10}\ Q}$$

E^{0}_{cell} = 0.029 V
n = 1

E_{cell}= 0.029V -(0.0592V/1)log $[Fe^{3+}]/[Fe^{2+}][Ag]$

=0.029V - 0.0592V*log [0.2]/[0.1]*[1.0]

=0.011V

"It decreases E_{cell} by 0.011 V (towards negative direction), hence, under these concentrations the reaction has lower tendency to form product".

Problem 3

a) What would happen to the E_{cell} of the previous example problem if the concentration of Fe^{2+} is 0.2 M and Fe^{3+} is 0.1 M?

b) Comment on the tendency of this reaction to form the product based on E_{cell}. How can you relate this observation with Le Chatelier's principle?

$$Zn\,(s) + Cu^{2+}\,(aq) \longrightarrow Zn^{2+}\,(aq) + Cu\,(s)$$

For this cell n = 2. If it is found that the cell potential is 1.10 V when Cu^{2+} and Zn^{2+} concentration is 1 mol/L. Then,

$$
\begin{aligned}
DG &= nFE \\
&= -\ (2 \text{ mol } e^{-}) \times 9.65 \times 10^{4}\ \text{C/mol } e^{-} \times 1.10\ \text{J/C} \\
&= -\ 2.12 \times 10^{5}\ \text{J} \quad = -\ 212\ \text{J}
\end{aligned}
$$

The ***standard cell potential*, E°**, of a galvanic cell is the cell potential measured when the concentration of each type of ion in the cell reaction is 1 mol/L (*standard state*).

The standard potential of this cell at 25°C is the potential when $[Cu^{2+}]$ and $[Zn^{2+}]$ are 1 mol/L. The cell potential is 1.10 V, and the ΔG^{o} is –212 J.

Problem 4

Assume that a galvanic cell reaction follows one electron transfer with a standard cell potential of 1.0 V. Determine the value of *K* for the overall cell reaction at 25 ^{o}C.

Solution:

$$E^\circ = \frac{0.0592\ \mathrm{V}}{n} \log K$$

$$K = 10^{\frac{(1)(1.0\ \mathrm{V})}{0.0592\ \mathrm{V}}} = 7.8 \times 10^{16}$$

Problem 5 **Calculating K and ΔG^o from E^o_{cell}**

PROBLEM Lead can displace silver from solution:

$$Pb(s) + 2Ag^+(aq) \longrightarrow Pb^{2+}(aq) + 2Ag(s)$$

As a consequence, silver is a valuable byproduct in the industrial extraction of lead from its ore. Calculate K and ΔG° at 298.15 K for this reaction.

Given: $E^0_{Ag+/Ag} = 0.80\ V$ $E^0_{Pb2+/Pb} = -\ 0.13\ V$

SOLUTION Writing the half-reactions with their E° values:

$$(1) \quad Ag^+(aq) + e^- \longrightarrow Ag(s) \qquad E^\circ = 0.80\ V$$

$$(2) \quad Pb^{2+}(aq) + 2e^- \longrightarrow Pb(s) \qquad E^\circ = -0.13\ V$$

Step 1 Calculation of E^0_{Cell}

$$E^0_{cell} = E^0_{cathode} - E^0_{Anode}$$

$$E^0_{cell} = E^0_{Ag+/Ag} - E^0_{Pb2+/Pb}$$
$$= 0.80 - (-0.13) = 0.93\ V$$

Anode = ?   Cathode = ?

Anode : Oxidation
Cathode : Reduction
Or
Higher the value of reduction potential will act as cathode

Step 2 Calculation of ΔG^0

$$\Delta G^0 = - nF E^0_{cell}$$

No of electron transfer (n) = 2
F = 965000 C

$$\Delta G^0 = - 2 \times 96500 \times 0.93$$
$$= 1.8 \times 10^2\ kJ/\ mol$$

Step 3 Calculation of K

$$E^\circ_{cell} = \frac{0.0592\ V}{n} \log K$$

$$\log K = \frac{nE^\circ_{cell}}{0.0592\ V}$$

$$\log K = \frac{0.93\ V \times 2}{0.0592\ V} = 31.42$$

$$K = 2.6 \times 10^{31}$$

Problem 6

$$Mg/Mg^{2+} \,||\, Zn^{2+}/Zn$$

What is the chemical reaction that takes place, and what is the standard free energy change for that reaction?

$E^0_{Mg2+/Mg}$ = -2.52 V $\qquad$ $E^0_{Zn2+/Zn}$ = - 0.76 V

Ans

$$Mg \longrightarrow Mg^{2+} + 2e-$$

$E^0_{Mg2+/Mg}$ = -2.52 V

$E^0_{Mg/Mg2+}$ = + 2.52 V

$$Zn^{2+} + 2e- \longrightarrow Zn$$

$E^0_{Zn2+/Zn}$ = - 0.76 V

Problem 7

Determine the oxidation-reduction reaction and calculate the standard cell potential in a galvanic cell composed of the aqueous ion/metal electrode pair: Zn/Zn^{2+} and Ag/Ag^{+}.

$E^0(Ag^+/Ag) = 0.80$ V $\quad\quad$ $E^0(Zn^{2+}/Zn) = -0.76$ V

Ans.

1. **Determine which half cell will function as the anode and which will function as the cathode.**
   The reduction half reactions for each half cell are:
   $Zn^{2+}(aq) + 2e^- \rightarrow Zn^0(s)$ standard reduction potential $= -0.76\,V$
   $Ag^+(aq) + 1e^- \rightarrow Ag^0(s)$ standard reduction potential $= +0.80\,V$
   The standard reduction potential of the Zn^{2+}/Zn half cell is more negative than the standard reduction potential of the Ag^0/Ag^+ half cell.
   Zinc will be the cathode and silver will be the anode.
2. **Balance the half reactions.**
   $Zn^{2+}(aq) + 2e^- \rightarrow Zn^0(s)$: gain of two electrons
   $Ag^+(aq) \rightarrow Ag^0(s) + 1e^-$: loss of one electron
   $2Ag^+(aq) \rightarrow 2Ag^0(s) + 2e^-$: loss of two electrons
3. **Write the balanced oxidation-reduction reaction for the galvanic cell.**
   $Zn^0(s) + 2Ag^+(aq) \rightarrow Zn^{2+} + 2Ag^0(s)$
4. **Determine the standard cell potential for the galvanic cell.**
   $E^0{}_{cell} = E^0{}_{cathode} - E^0{}_{anode} = 0.80V - (-0.76V) = 1.56V$

Note that while the silver half reaction is multiplied by 2 to get the balanced oxidation-reduction equation, ***the standard reduction potential for the half cell reaction is not changed*** when calculating the potential for the half cell pair.

Problem 8

What is the potential of a galvanic cell with the redox couples: Cd^{2+}/Cd and Pb^{2+}/Pb, at 25°C with the concentrations of Cd^{2+} at 0.02 M and Pb^{2+} at 0.20 M?

$E^0(Cd^{2+}/Cd) = -0.40\ V$ $\quad$ $E^0(Pb^{2+}/Pb) = -0.13\ V$

Ans.

1. **Determine the overall cell reaction.**
   Since both half reactions involve the transfer of two electrons, the overall reaction is:

   $$Pb^{2+}(aq) + Cd(s) \rightarrow Cd^{2+}(aq) + Pb(s)$$

2. **Determine the cell potential under standard conditions.**

   E^0 of $Cd^{2+} = -0.40$; E^0 of Pb^{2+} is -0.13.

   Cadmium will be the anode since it has the lowest reduction potential and lead will be the cathode.
   The standard cell potential is:

   $$E^0{}_{cell} = E^0{}_{cathode} - \mathrm{E}^0{}_{anode} = -0.13 - (-0.40) = 0.27\ \mathrm{V}$$

3. **Determine the cell potential with the nonstandard concentrations.**

   $$E_{cell} = E^0{}_{cell} - \frac{0.059}{n^{e-}} \log Q = 0.27\ \mathrm{V} - \frac{0.059}{2}\ \log\ (0.02\mathrm{M})/(0.20\mathrm{M})$$

   $$E_{cell} = 0.27\ \mathrm{V} - 0.03\ \log\ (0.10) = 0.27 + 0.03 = 0.3\ \mathrm{V}$$

Problem 9

What is the cell potential of a galvanic cell composed of two Ag/Ag^{+} half cells, one with a silver ion concentration of 0.01 M and the other with a silver ion concentration of 1M?

$E^{0}(Ag^{+}/Ag) = 0.80\ V$

Ans

1. **Determine the oxidation-reduction reaction.**

The cell with the lower concentration (0.01 M) becomes the anode:

$$Ag^0(s) \rightarrow Ag^+(aq)\,[0.01\,M] + 1e^- \quad \text{(anode)}$$

The cell with the higher concentration becomes the cathode:

$$Ag^+(aq)\ [1.0\,M] + 1e^- \rightarrow Ag^0(s) \quad \text{(cathode)}$$

The chemical equation for the oxidation-reduction reaction is:

$$Ag^0(s) + Ag^+[1.0\,M] \rightarrow Ag^+[0.01\,M] + Ag^0(s)$$

2. **Determine the cell potential.**

$$E = -\frac{0.059}{n^{e-}} \log Q$$

where: $E^0 = 0$, $n^{e-} = 1$, $Q = [0.01\,M]/[1.0\,M]$

$$\begin{aligned} E &= E^0 - 0.059 \times \log \left[Ag^+ 0.01M\right]/\left[Ag^+ 1.0M\right] \\ &= 0\ V - [(0.059)(-2)] \\ &= 0.118\ V \end{aligned}$$

Cell Potential and pH

$$E_{cell} = E^o_{cell} - \frac{0.0592}{n} \log_{10} Q$$

What is the pH in a cell in which Mg^{2+} = 1 M and pressure of H_2 = 1 atm, If the E_{cell} is 2.099 V ?
[Given: $E^0(Mg^{2+}/Mg)$ = - 2.363 V]

Ans. $Mg + 2\,H^+ \rightleftharpoons Mg^{2+} + H_2$

$$K = \frac{[Mg^{2^+}]}{[H+]^2} = \frac{1}{[H+]^2}$$

$$E_{cell} = E^0_{cell} - \frac{0.0592}{2} log\ \frac{1}{[H+]^2} \qquad = 2.363 - 0.0592\, pH$$

$$p^H = 4.46$$

Cell Potential and pH

$H_2\,(g) \rightarrow 2\,H^+\,(1\,M) + 2e-$ (Anode)

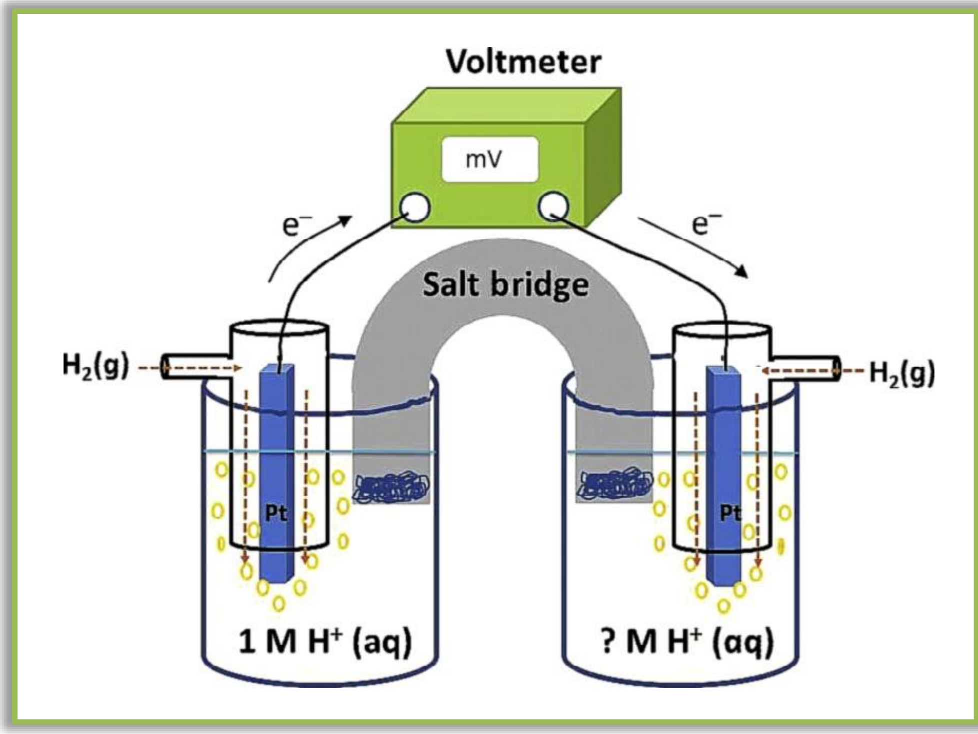
$2\,H^+\,(x\,M) + 2e- \rightarrow H_2\,(g)$ (Cathode)

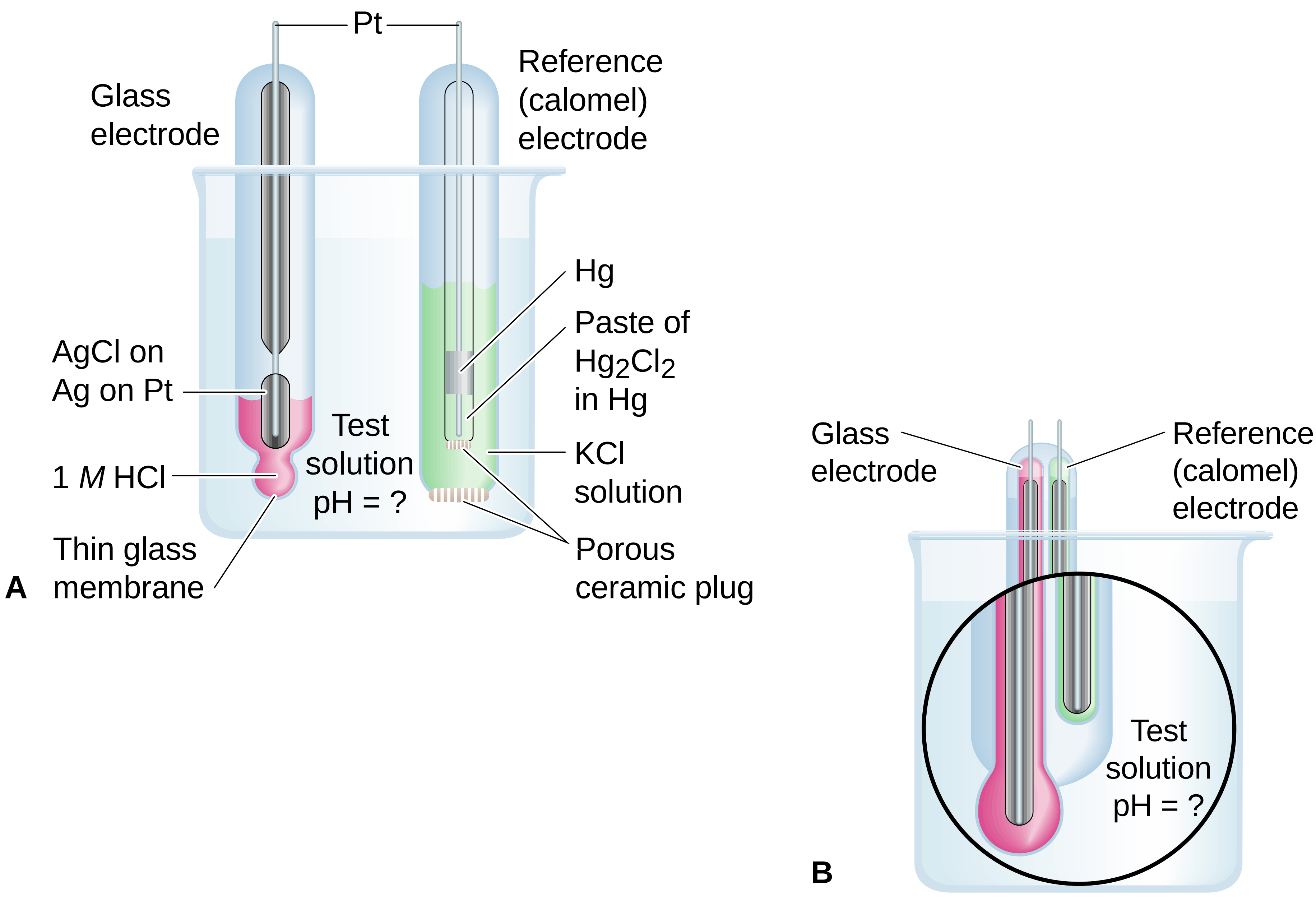
$$\boxed{E_{cell} = E^o_{cell} - \frac{0.0592}{n}\log_{10}\,Q}$$

$$K = \frac{1}{[H+]^2}$$

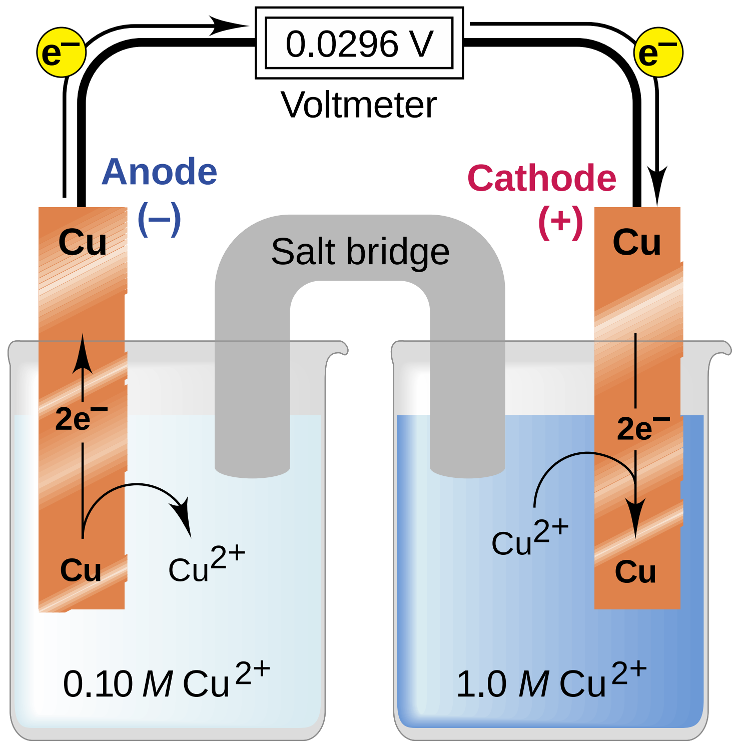
$$E_{cell} = E^0_{cell} - \frac{0.0592}{2}\,log\,\frac{1}{[H+]^2}$$

$$\mathbf{E\,(2H^+/H_2) = -0.0592\,pH}$$




Pt
Glass electrode
Reference (calomel) electrode
Hg
Paste of Hg_2Cl_2 in Hg
AgCl on Ag on Pt
Test solution pH = ?
KCl solution
1 *M* HCl
Thin glass membrane
Porous ceramic plug
A
Glass electrode
Reference (calomel) electrode
Test solution pH = ?
B

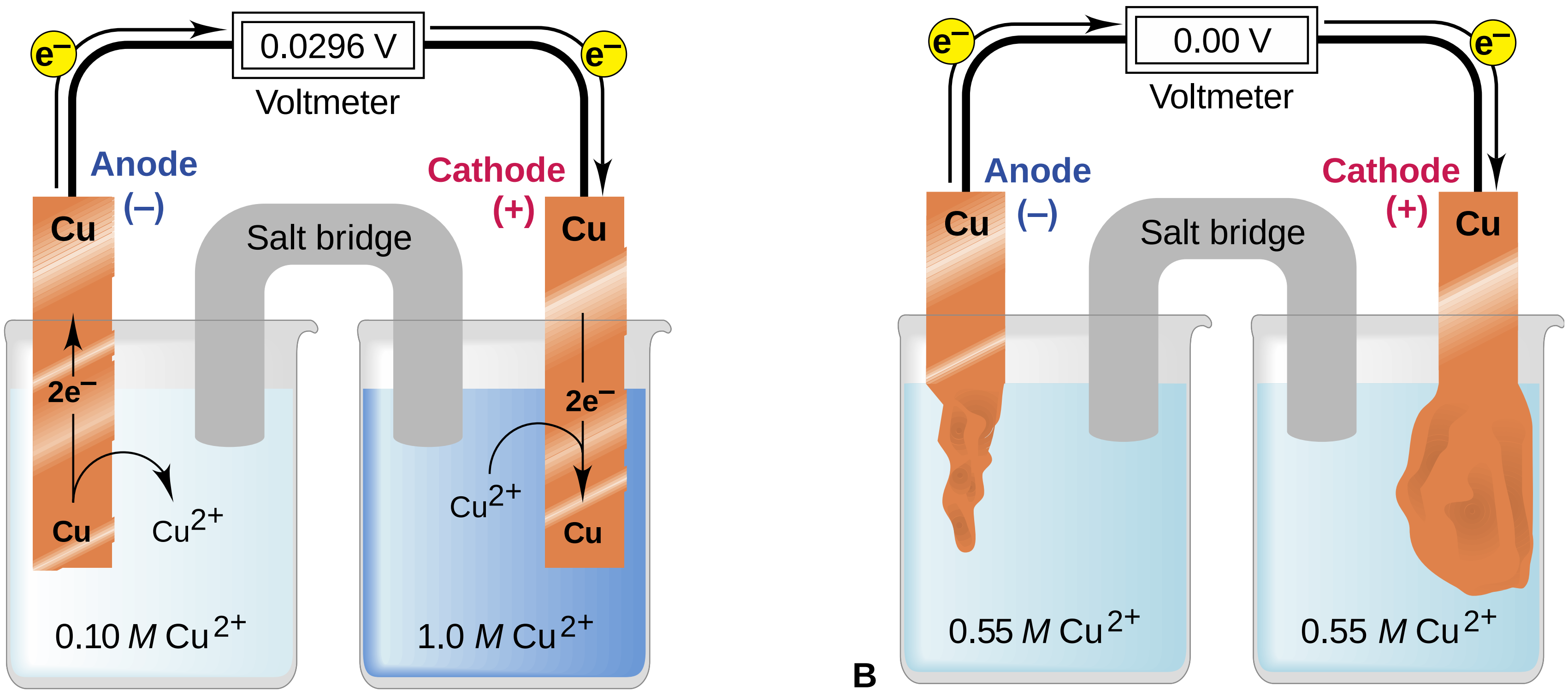
Concentration Cells



0.0296 V
Voltmeter
e⁻
e⁻
Anode
(–)
Cathode
(+)
Cu
Cu
Salt bridge
2e⁻
2e⁻
Cu
Cu²⁺
Cu²⁺
Cu
0.10 M Cu²⁺
1.0 M Cu²⁺

What is concentration cell?

- Concentration cell is a limited form of galvanic cell that has *two equivalent half-cells,* only the concentration of ions are different in half-cells.



Oxidation half-reaction

$Cu(s) \longrightarrow Cu^{2+}(aq;\ 0.10\ M) + 2e^-$

Reduction half-reaction

$Cu^{2+}(aq;\ 1.0\ M) + 2e^- \longrightarrow Cu(s)$

Overall (cell) reaction

$Cu^{2+}(aq;\ 1.0\ M) \longrightarrow Cu^{2+}(aq;\ 0.10\ M)$

What is the standard cell potential of concentration cell?

E^0_{Cell} = E^o(Cathode) – E^o(Anode)

Since, equivalent half-cells are used in concentration cells

$$E^0_{Cell} = X - X = 0$$

Therefore, E_{cell} depends only on difference in concentration of ions. In cathodic and anodic compartments.
At 25 ^oC, E_{cell} is expressed by simplified Nernst equation in which E^0_{Cell} = 0.

$$E = -\frac{0.059}{n_{e^-}} \log Q$$

What is the cell potential of a galvanic cell composed of two Ag/Ag^{+} half cells, one with a silver ion concentration of 0.01M and the other with a silver ion concentration of 1M?

1. Determine the oxidation-reduction reaction.

The cell with the lower concentration (0.01 M) becomes the anode:

$$Ag^{0}(s) \rightarrow Ag^{+}(aq)\,[0.01\,M] + 1e^{-} \quad \text{(anode)}$$

The cell with the higher concentration becomes the cathode:

$$Ag^{+}(aq)\ [1.0\,M] + 1e^{-} \rightarrow Ag^{0}(s) \quad \text{(cathode)}$$

The chemical equation for the oxidation-reduction reaction is:

$$Ag^{0}(s) + Ag^{+}[1.0\,M] \rightarrow Ag^{+}[0.01\,M] + Ag^{0}(s)$$

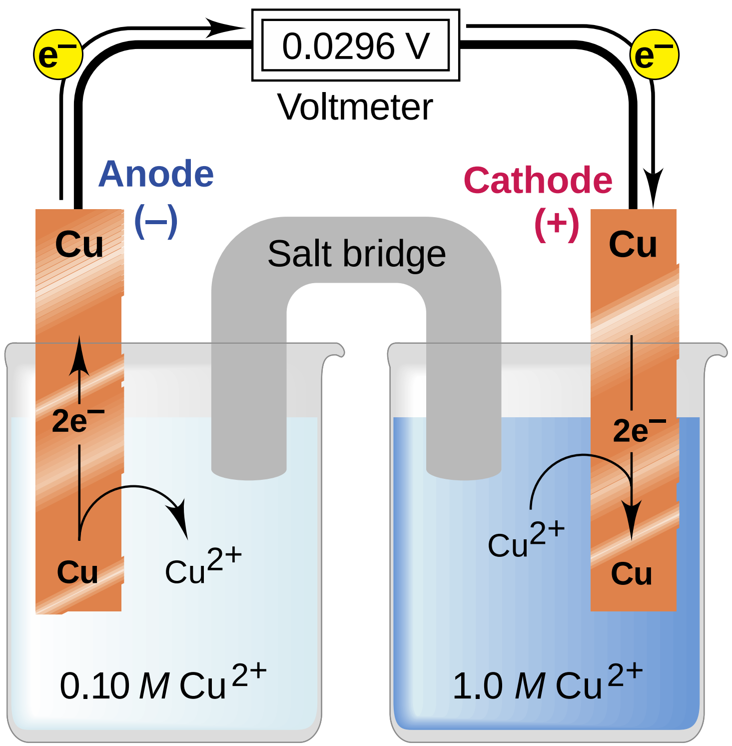
2. Determine the cell potential.

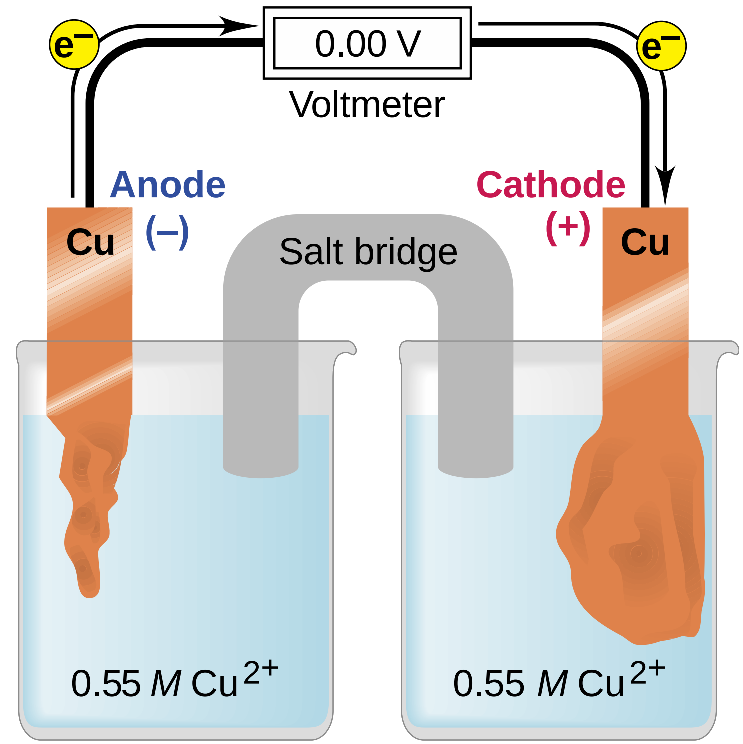
$$E = -\frac{0.059}{n^{e-}} \log Q$$

where: $E^{0} = 0$, $n^{e-} = 1$, $Q = [0.01\,M]/[1.0\,M]$

$$\begin{aligned} E &= E^{0} - 0.059 \times \log \left[Ag^{+}0.01M\right]/\left[Ag^{+}1.0M\right] \\ &= 0\ V - [(0.059)(-2)] \\ &= 0.118\ V \end{aligned}$$

E^{o}_{Cell} for concentration cell = 0





B

Oxidation half-reaction

$Cu(s) \longrightarrow Cu^{2+}(aq;\ 0.10\ M) + 2e^-$

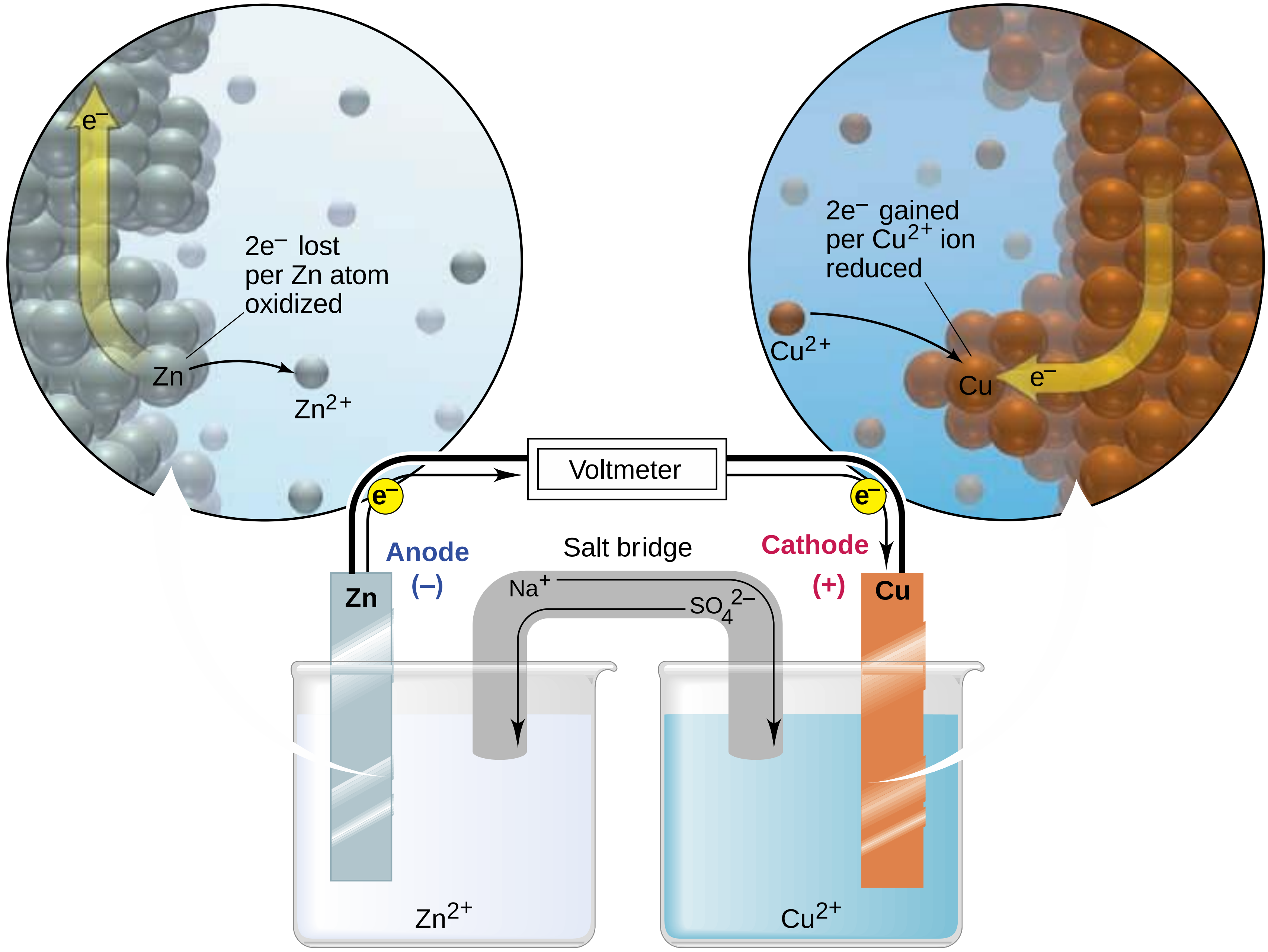
Reduction half-reaction

$Cu^{2+}(aq;\ 1.0\ M) + 2e^- \longrightarrow Cu(s)$

Overall (cell) reaction

$Cu^{2+}(aq;\ 1.0\ M) \longrightarrow Cu^{2+}(aq;\ 0.10\ M)$

$$E_{cell} = E^o_{cell} - \frac{0.0592}{n}\log_{10}\ Q$$



e⁻
2e⁻ lost
per Zn atom
oxidized
Zn
Zn2+
2e⁻ gained
per Cu2+ ion
reduced
Cu2+
Cu
e⁻
Voltmeter
e⁻
e⁻
Anode
(–)
Salt bridge
Cathode
(+)
Zn
Na+
SO4 2–
Cu
Zn2+
Cu2+

Oxidation half-reaction

$$Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^-$$

Reduction half-reaction

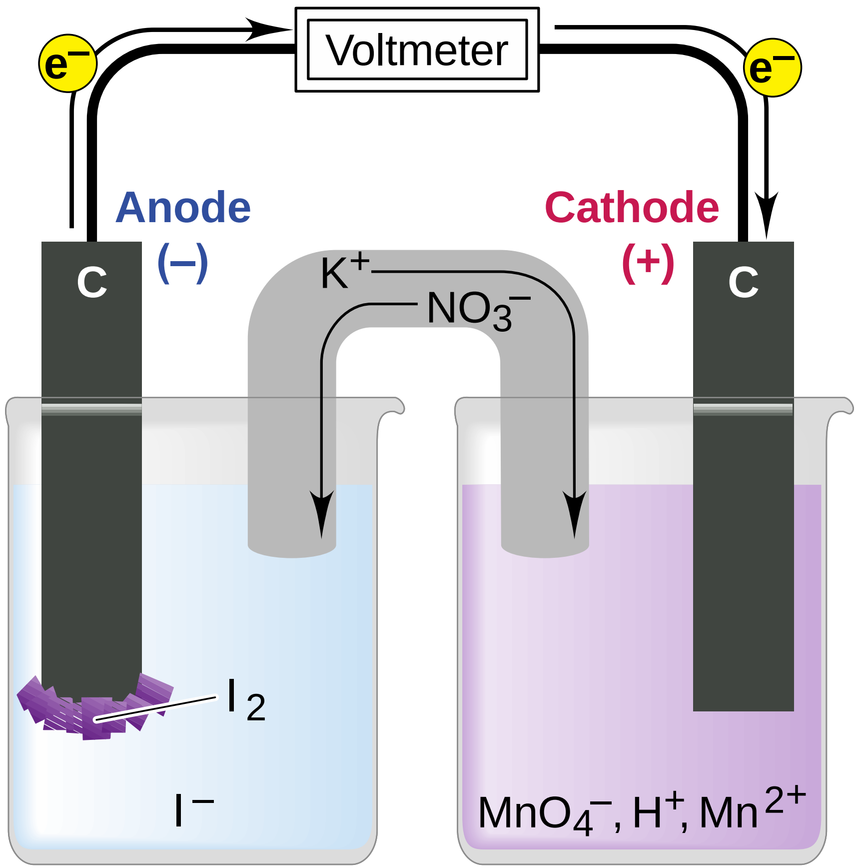
$$Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$$

Overall (cell) reaction

$$Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$$

$2I^-(aq) \pm £ \; I_2(s) + 2e^-$ [anode; oxidation]

$MnO_4^-(aq) + 8H^+(aq) + 5e^- \pm £ \; Mn^{2+}(aq) + 4H_2O(l)$ [cathode; reduction]



Oxidation half-reaction

$2I^-(aq) \longrightarrow I_2(s) + 2e^-$

Reduction half-reaction

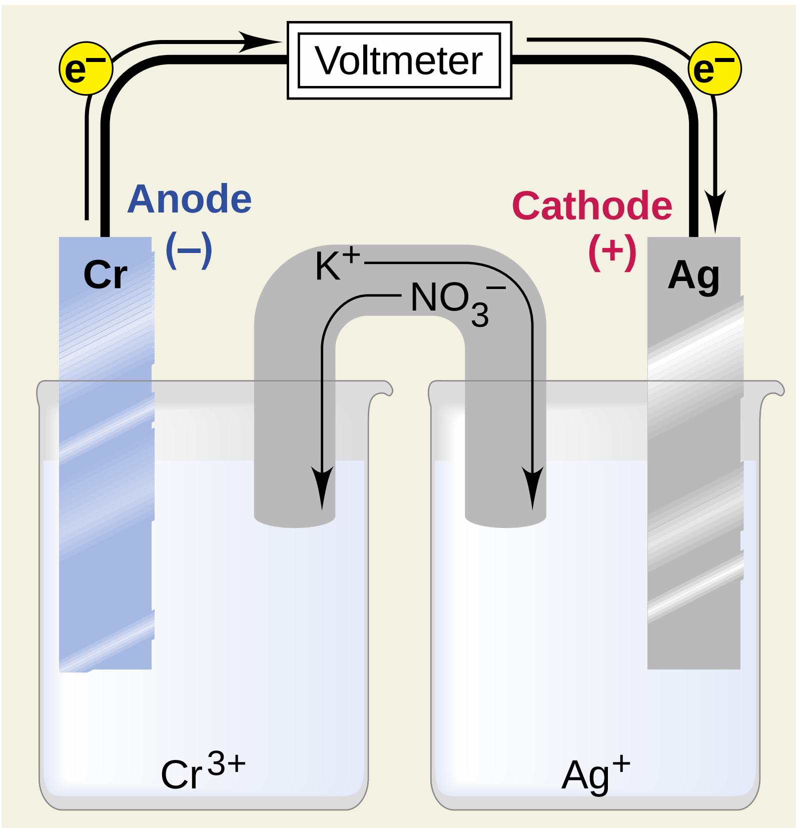
$MnO_4^-(aq) + 8H^+(aq) + 5e^- \longrightarrow Mn^{2+}(aq) + 4H_2O(l)$

Overall (cell) reaction

$2MnO_4^-(aq) + 16H^+(aq) + 10I^-(aq) \longrightarrow 2Mn^{2+}(aq) + 5I_2(s) + 8H_2O(l)$

graphite 0 $I^-(aq) \mid I_2(s)$ 00 $H^+(aq)$, $MnO_4^-(aq)$, $Mn^{2+}(aq)$ 0 graphite

$$Cr(s)\ 0Cr^{3+}(aq)\ 00Ag^{+}(aq)\ 0Ag(s)$$



Oxidation half-reaction

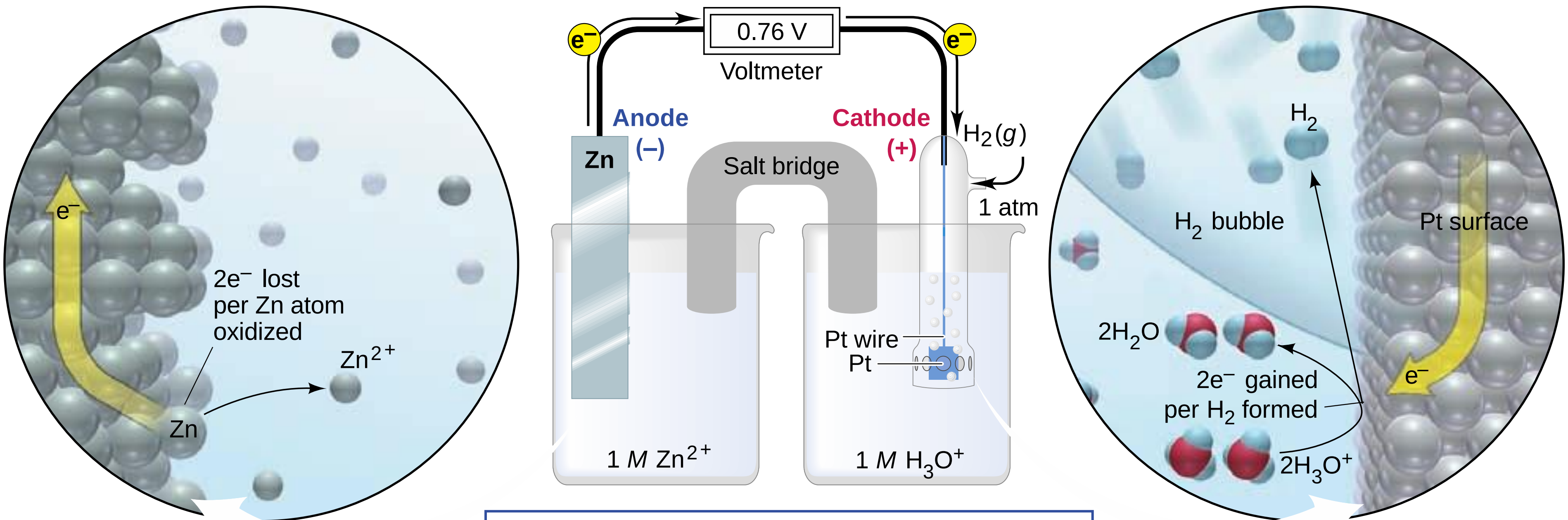
$Cr(s) \longrightarrow Cr^{3+}(aq) + 3e^{-}$

Reduction half-reaction

$Ag^{+}(aq) + e^{-} \longrightarrow Ag(s)$

Overall (cell) reaction

$Cr(s) + 3Ag^{+}(aq) \longrightarrow Cr^{3+}(aq) + 3Ag(s)$



Oxidation half-reaction

$Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^-$

Reduction half-reaction

$2H_3O^+(aq) + 2e^- \longrightarrow H_2(g) + 2H_2O(l)$

Overall (cell) reaction

$Zn(s) + 2H_3O^+(aq) \longrightarrow Zn^{2+}(aq) + H_2(g) + 2H_2O(l)$